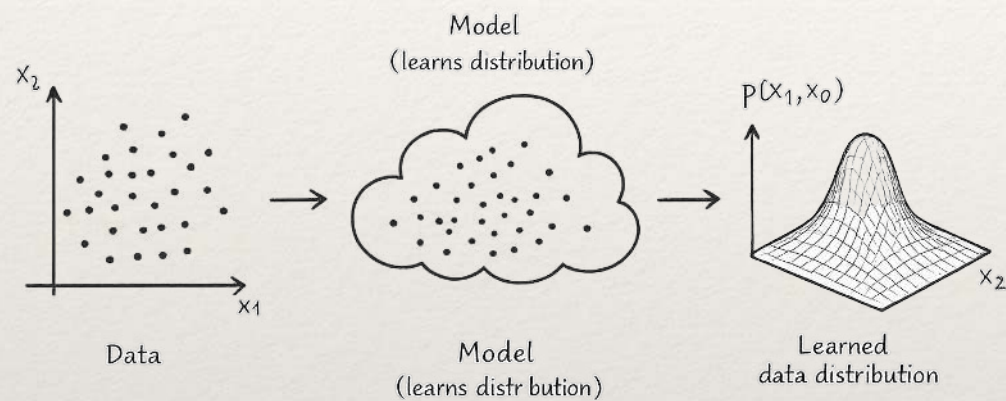

Physics-informed machine learning: when data does not tell the whole story

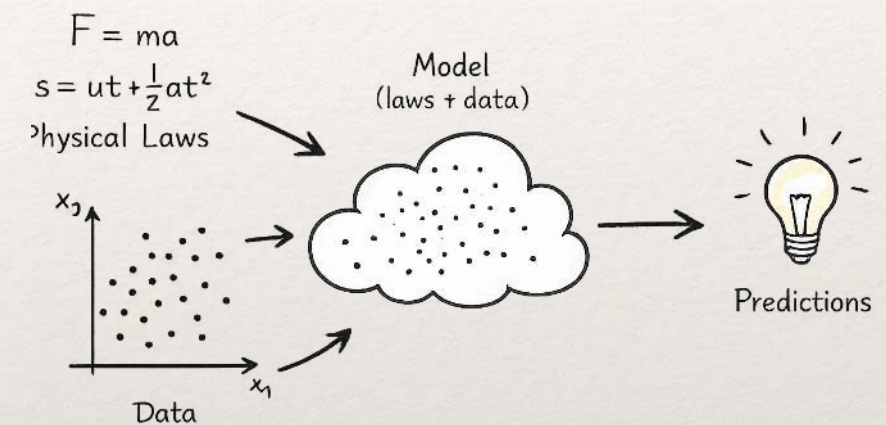
Dr. Suchita Kulkarni

Forefronts of AI/ML

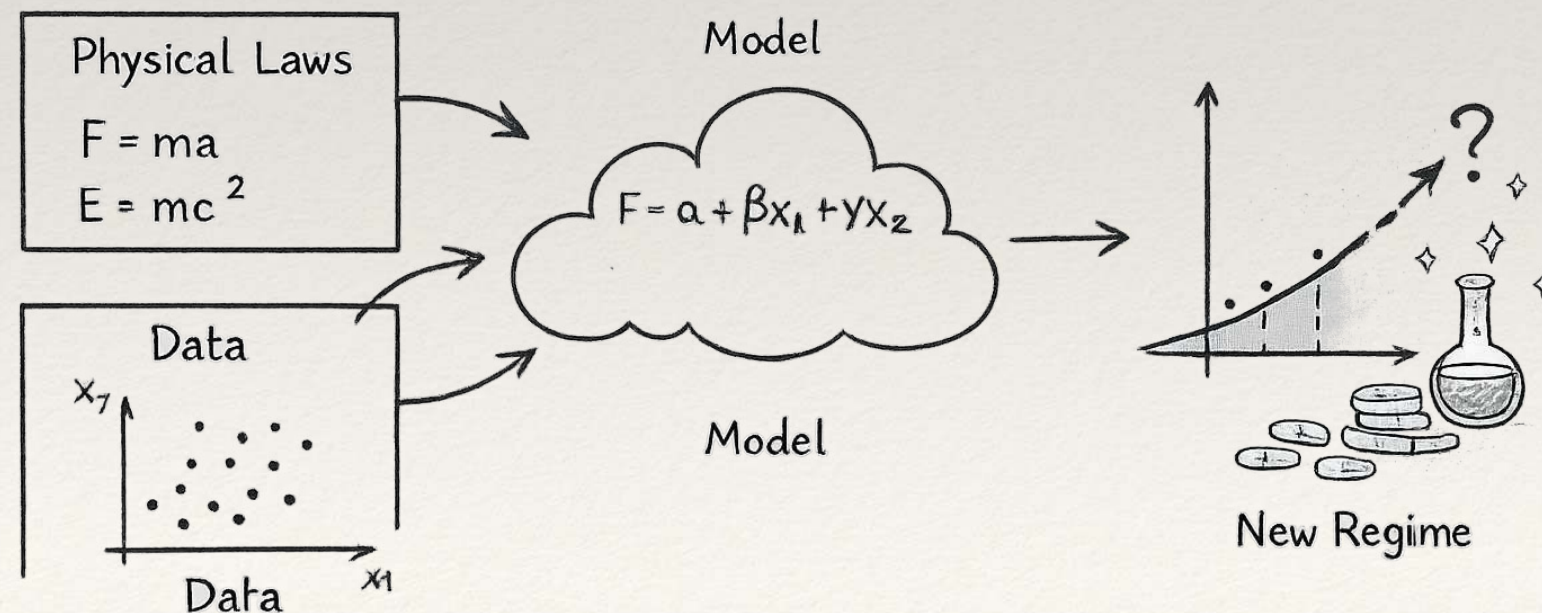
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Advancing understanding of the world for reliable predictions



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Optimization of Blast Furnace Ironmaking Using Machine Learning and Genetic Algorithms

Manendra Singh Parihar, Sri Harsha Nistala, Rajan Kumar, Sristy Raj, Adity Ganguly, Venkataramana Runkana

First published: 22 August 2024 | <https://doi.org/10.1002/srin.202300788> | [VIEW METRICS](#)

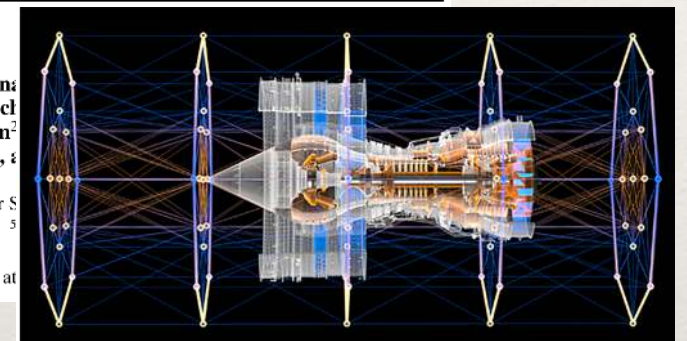
Advancing understanding of the world for reliable predictions

AURORA: A FOUNDATION MODEL OF THE ATMOSPHERE

Cristian Bodnar^{*,1}, Wessel P. Bruinsma^{*,1}, Anna
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¹Microsoft Research AI for S
²Microsoft Corporation ³JKU Linz ⁴University of Cambridge ⁵

^{*}Equal contribution [†]Work done while at



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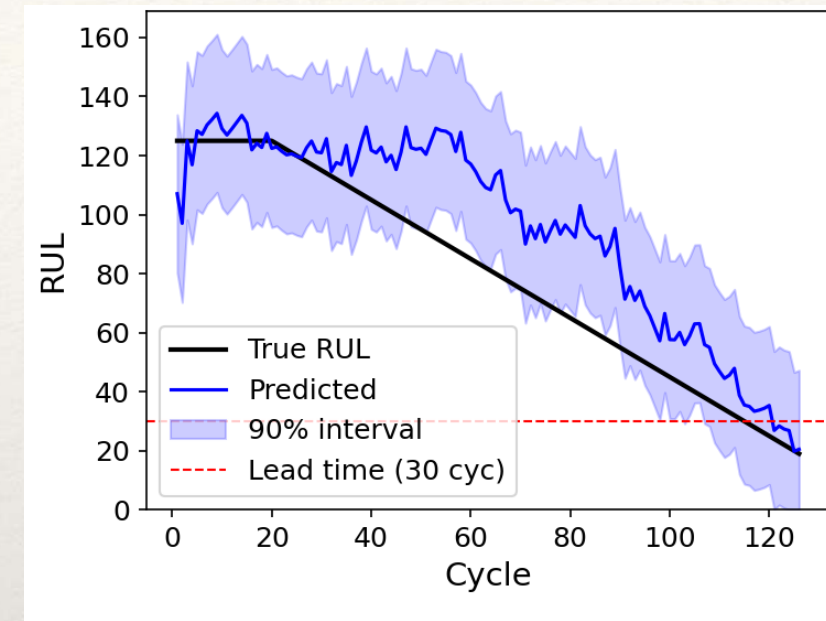
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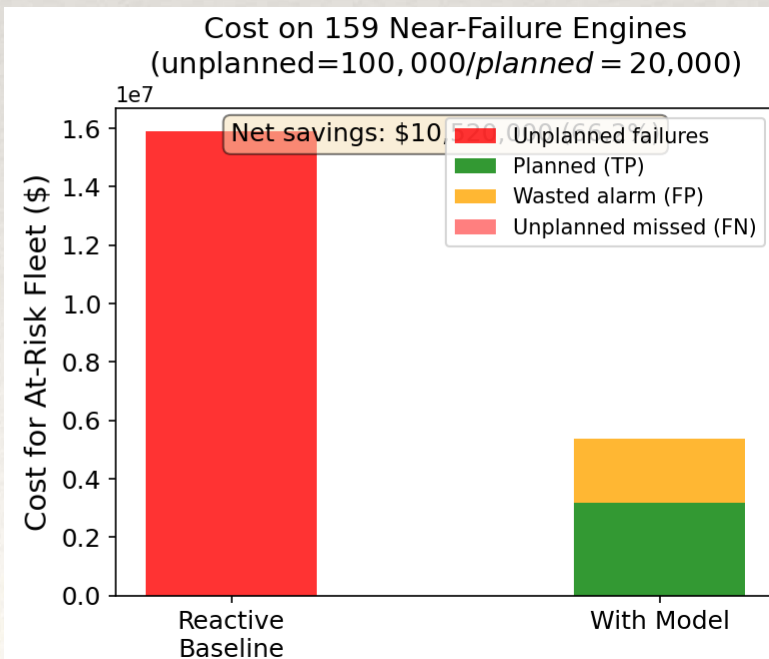
Jan 15, 2026

When surprises are expensive



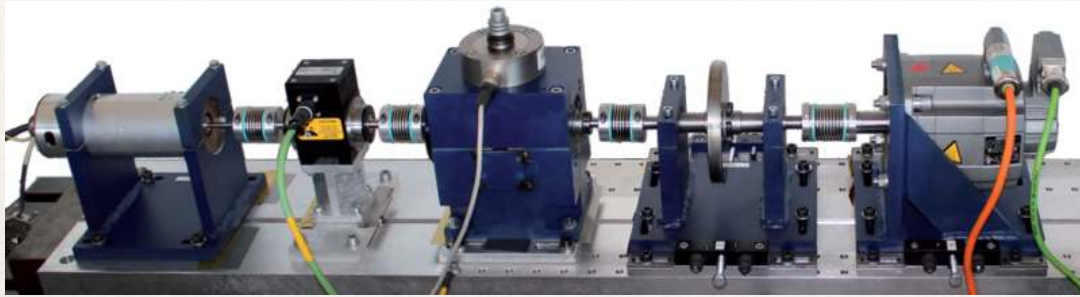
Detection Confusion Matrix (threshold=30 cycles)
Recall=100.0% Precision=59.1% Specificity=79.9%

		Alarm Fired	
		Yes	No
Near failure	TP (caught near-failure)	159	0 FN (missed near-failure)
	FP (false alarm)	110	438 TN (correctly quiet)



Maintenance planning reduces operational costs drastically

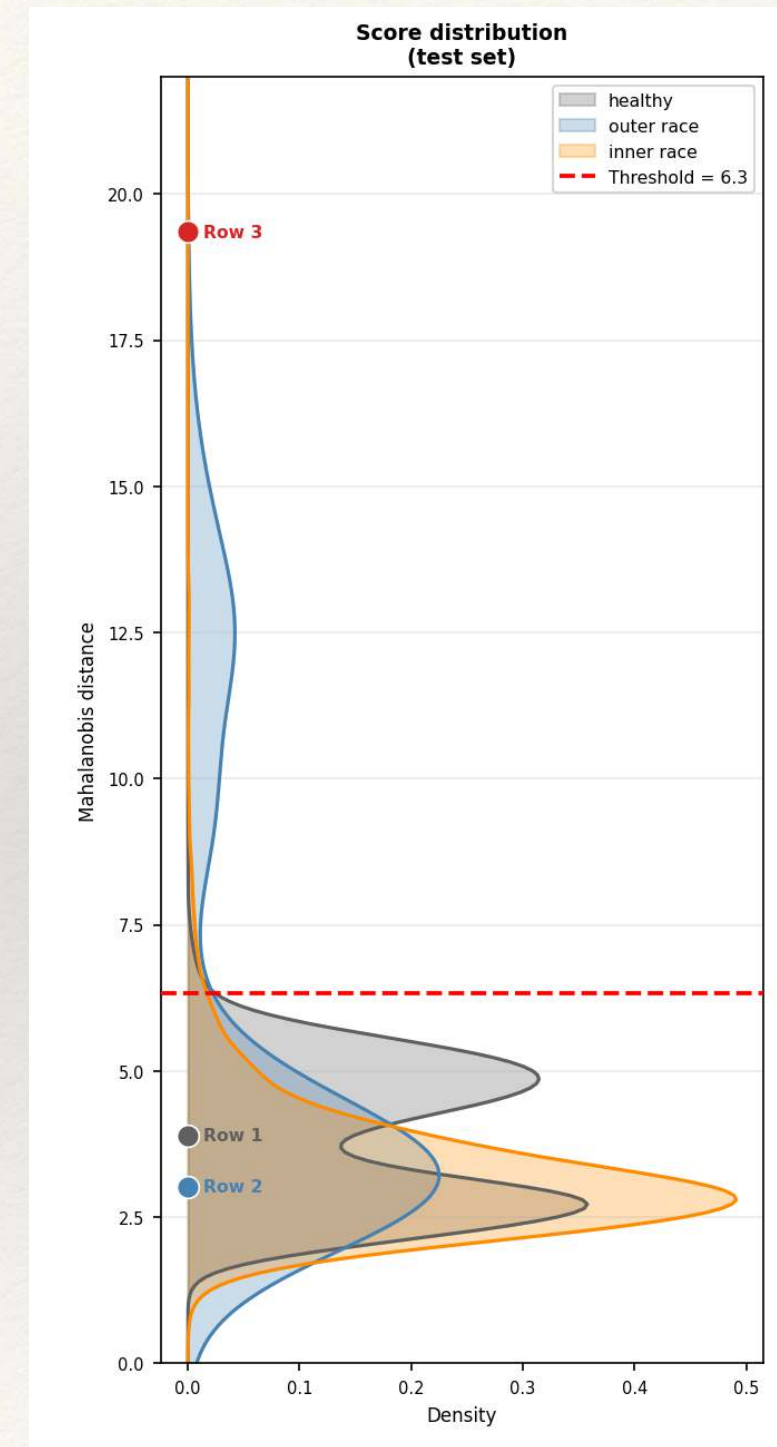
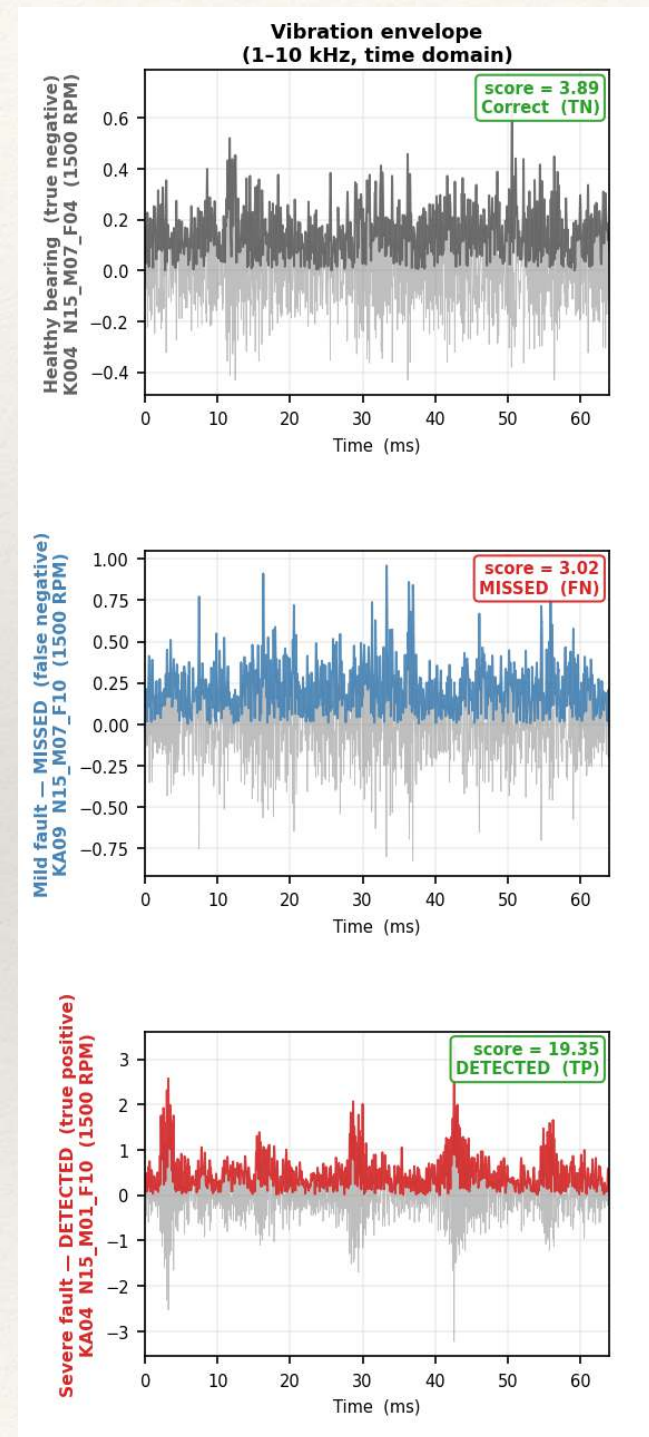
When traditional physics and ML doesn't help



Physics predicts how certain bearing faults should change vibration frequencies

Extract physics features and cluster
no NN necessary

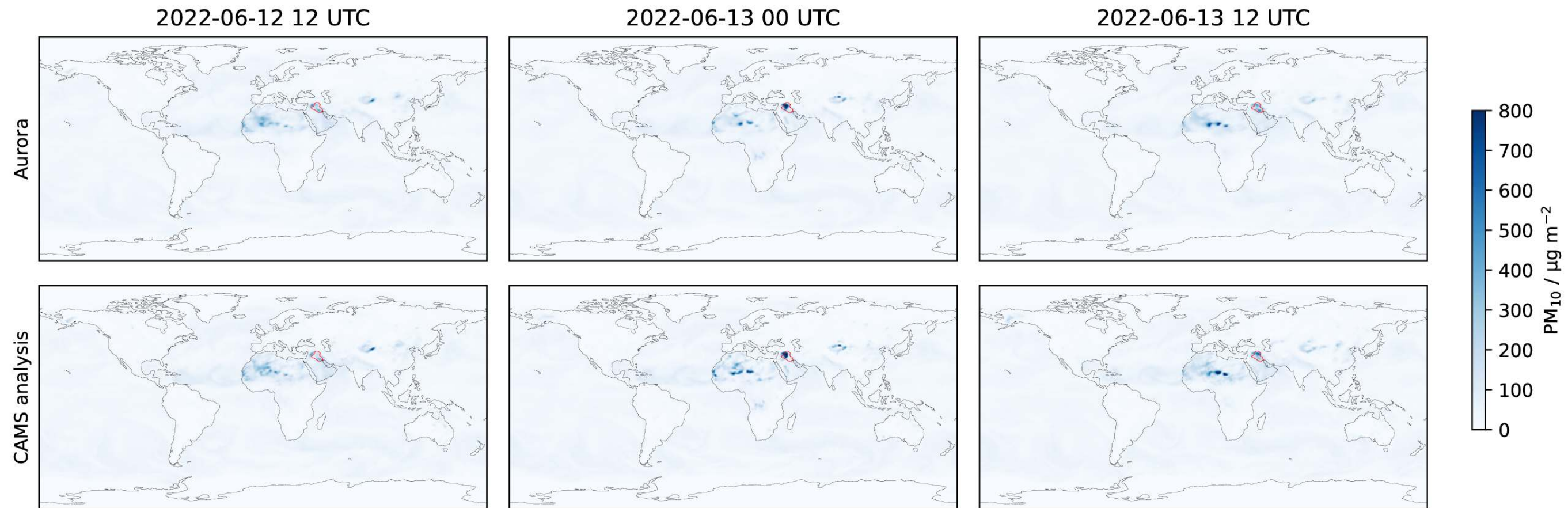
Problem: Pure physics feature extraction
and clustering does not separate subtle
anomalies



When extrapolations need principles

Aurora sees storm brewing a day in advance

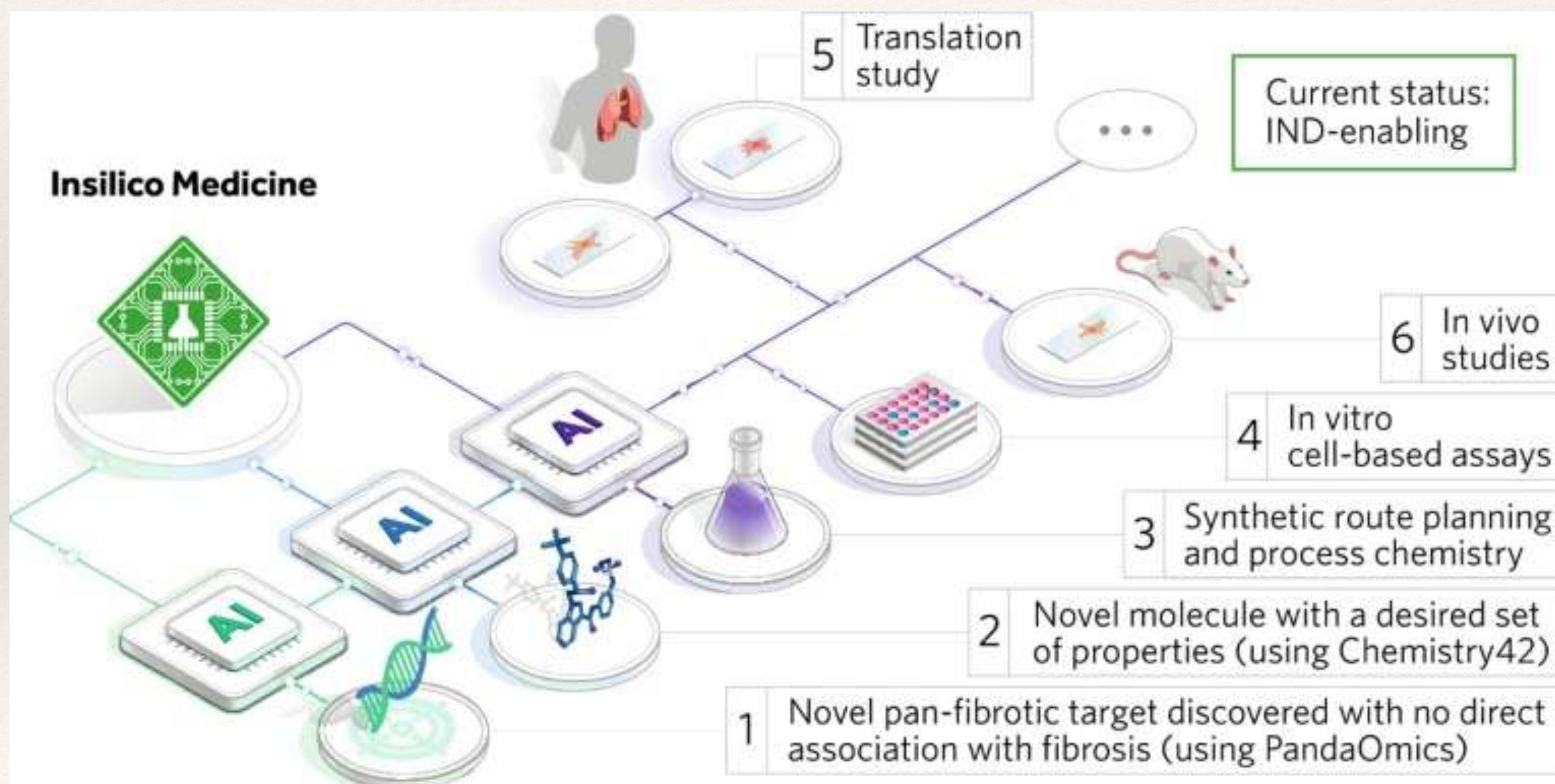
arXiv:2405.13063



Both Aurora and CAMS initialised at 12 Jun 2022 00 UTC

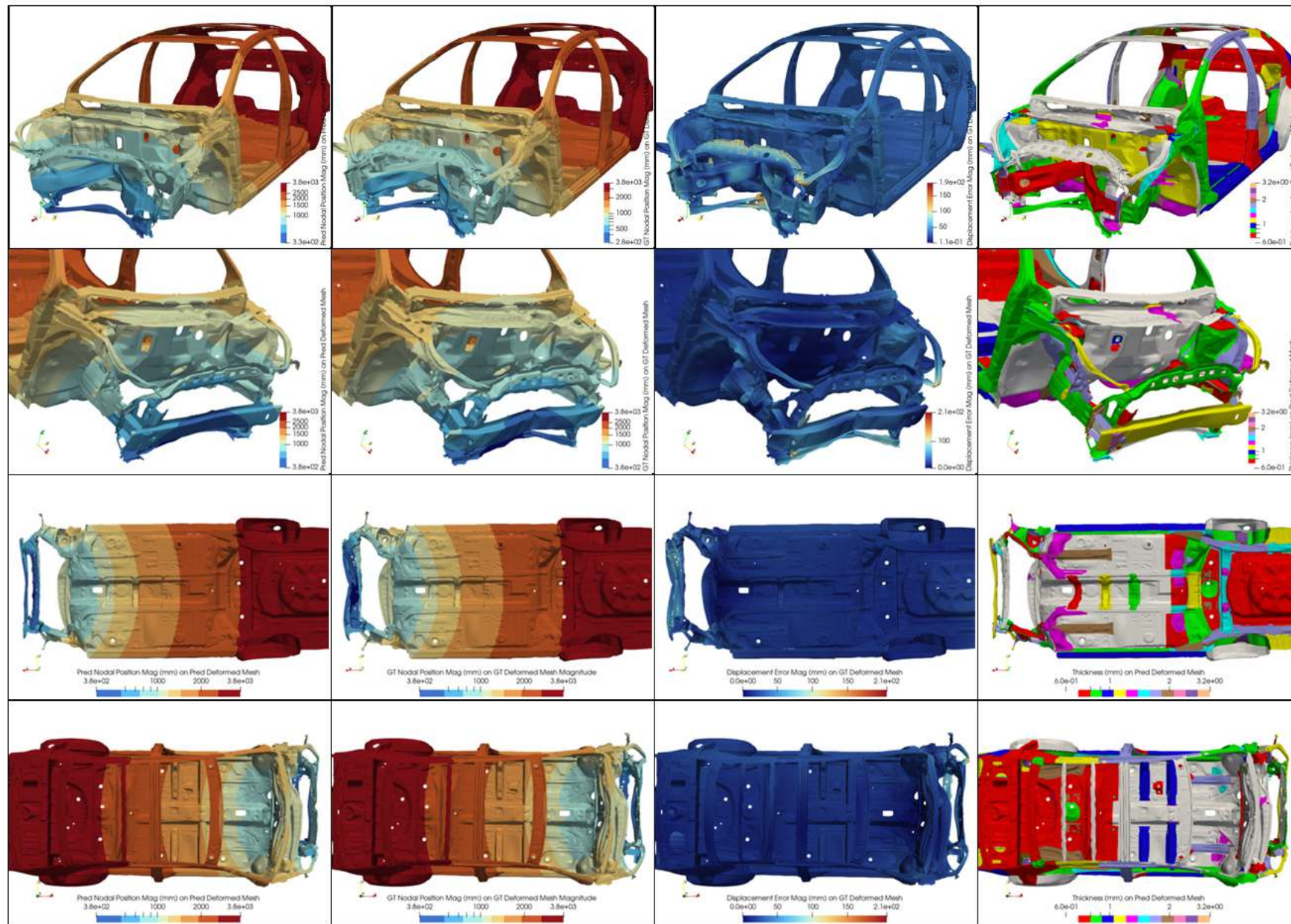
When discoveries are life-changing

From start to phase 1 in 30 months for pulmonary fibrosis drug development



When data is expensive

arXiv:2510.15201



HPC simulation

NN simulation

Difference

Damage thickness

- Physical crash test can cost between \$50k - \$600k
- Full numerical solver may take more than 15+ hours on a HPC
- AI/ML assisted crash testing reduces inference time to seconds and introduces smarter physical crash test strategies (training the model remains expensive)

Today's focus

Making the best decisions given real-life data

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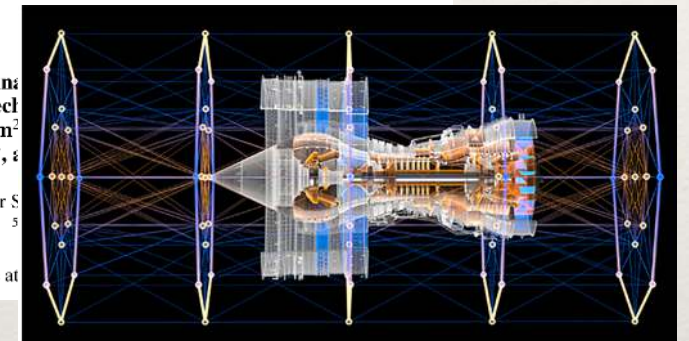
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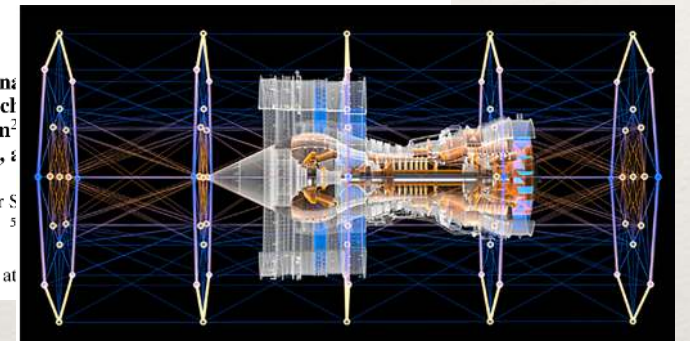
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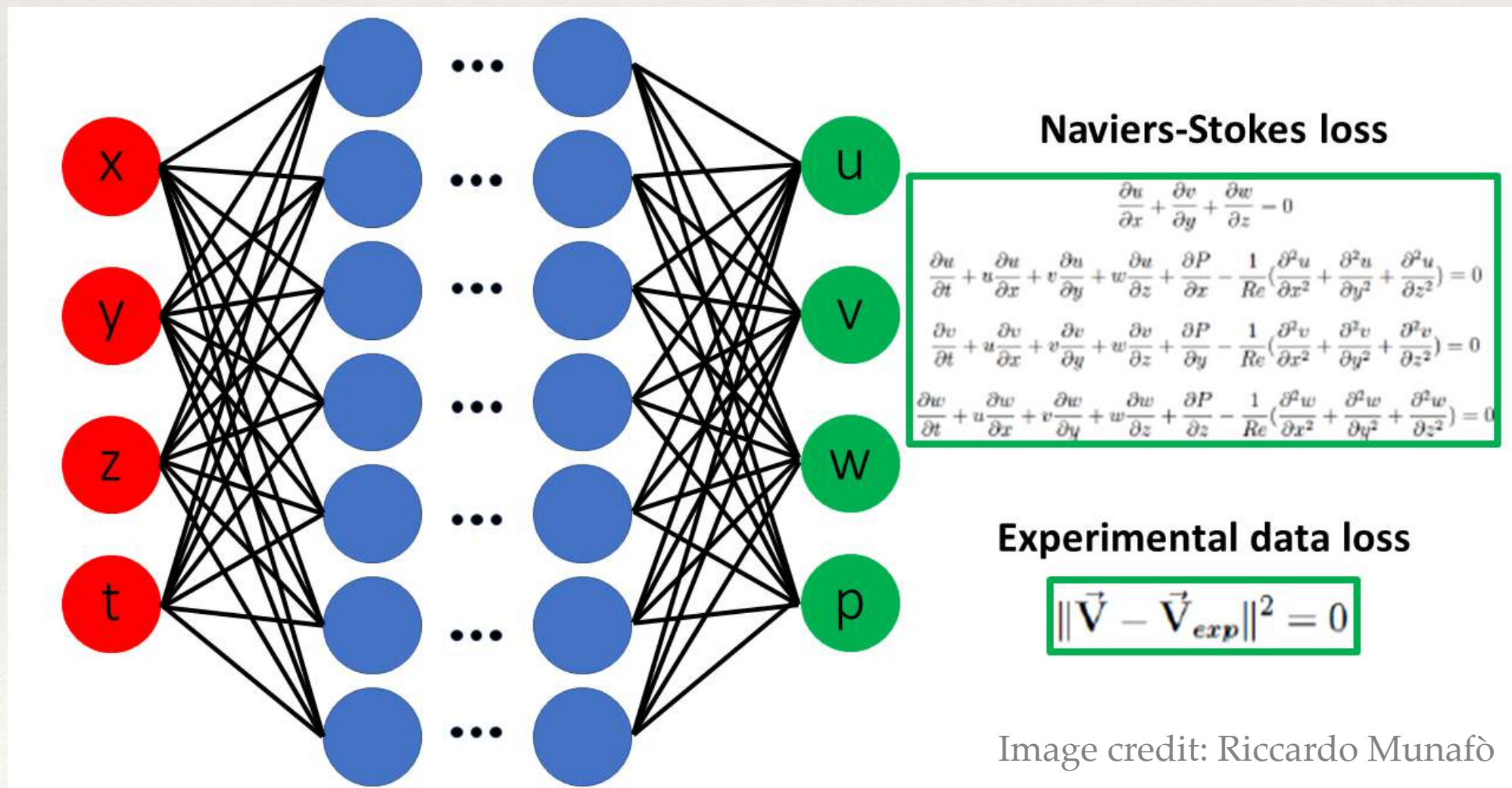
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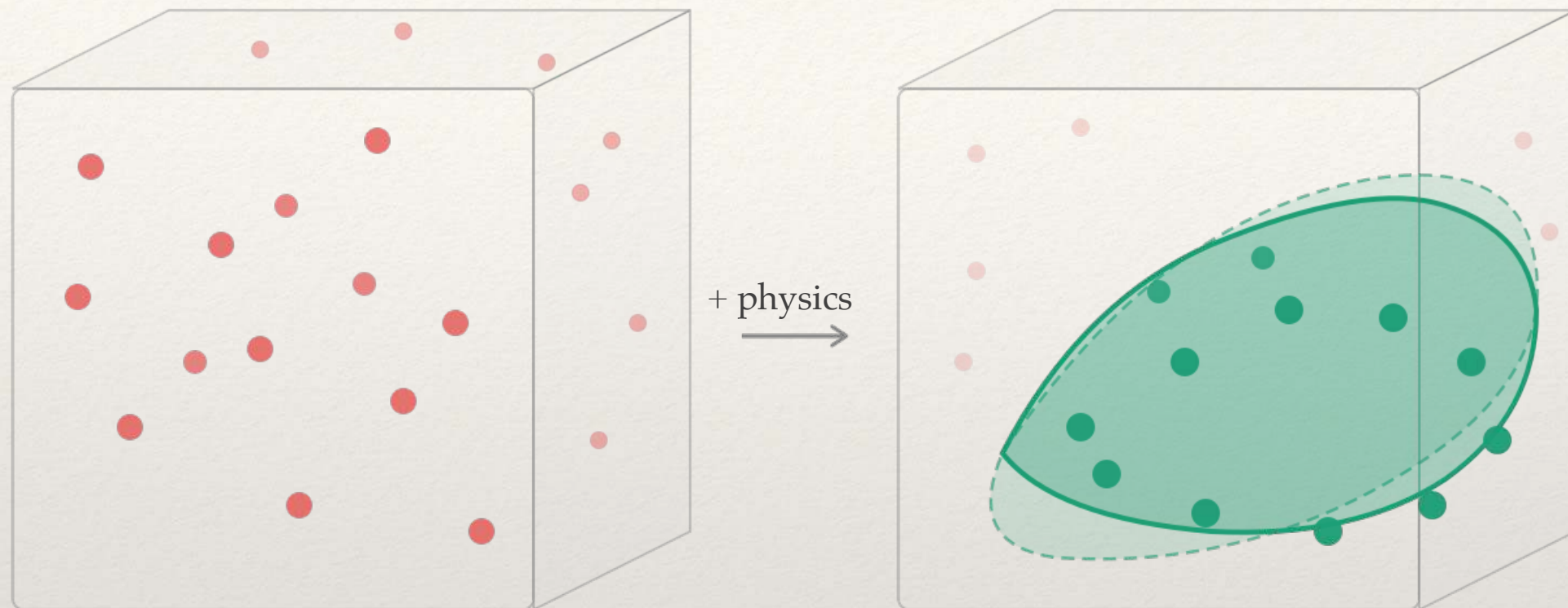


What is physics informed machine learning?

- Physics informed machine learning (PIML)
 - Art of injecting known knowledge of physical systems into machine learning
 - Suffers less from data biases, and extrapolates better
 - Contains built-in explainability



Why does physics help?



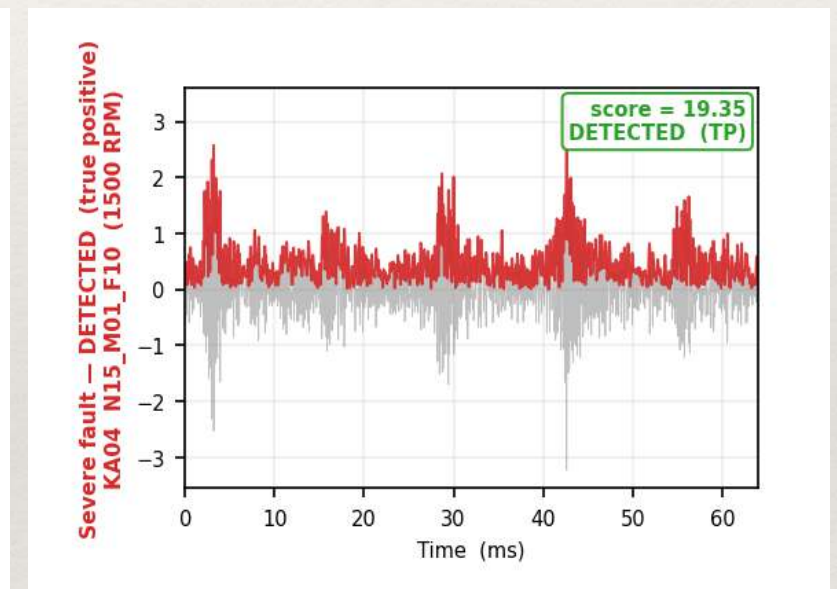
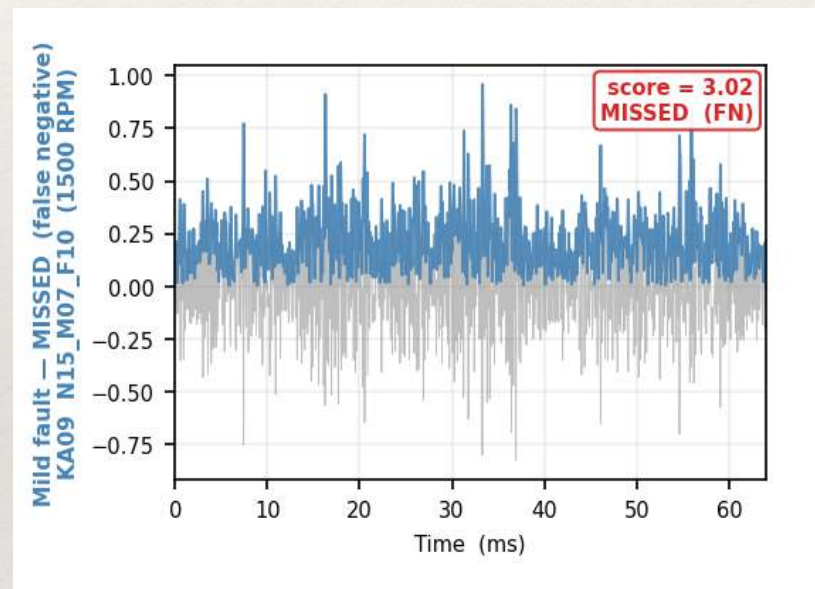
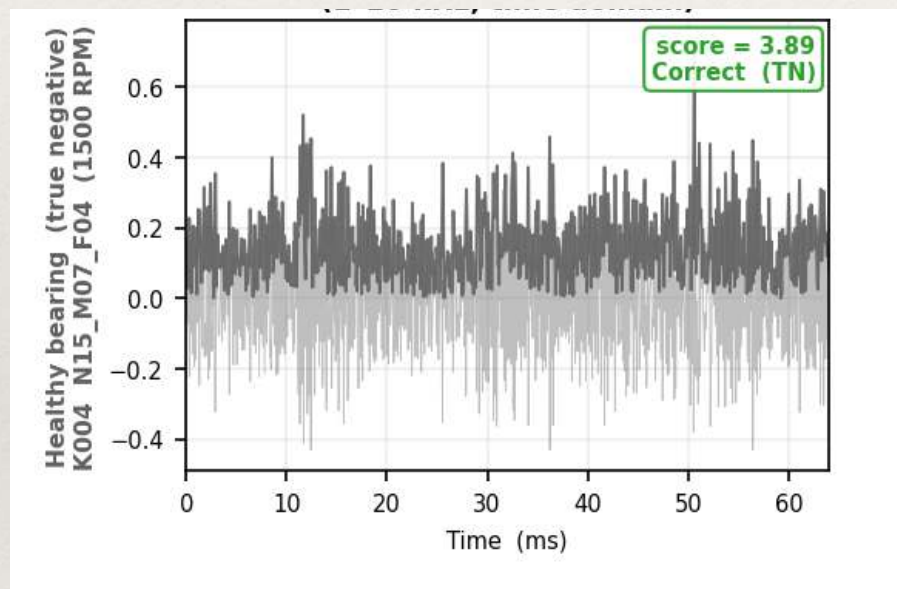
- Physics constraints restrict the space of possible solutions
 - The simple harmonic oscillator differential equation can have any amplitude or phase but every pair of y, t must satisfy the equation

$$\mathcal{L}_{phy} = \frac{d^2 y_{recon}}{dt^2} + \omega^2 y_{recon}$$

- Wrong physics model misidentifies solution space e.g. simple harmonic oscillator is a Taylor expansion in disguise, we will see general case later in the talk)

This is easier said than done

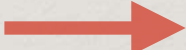
- Real life data is hardly ever clean
- Real life data is incomplete or sparse or anomalous



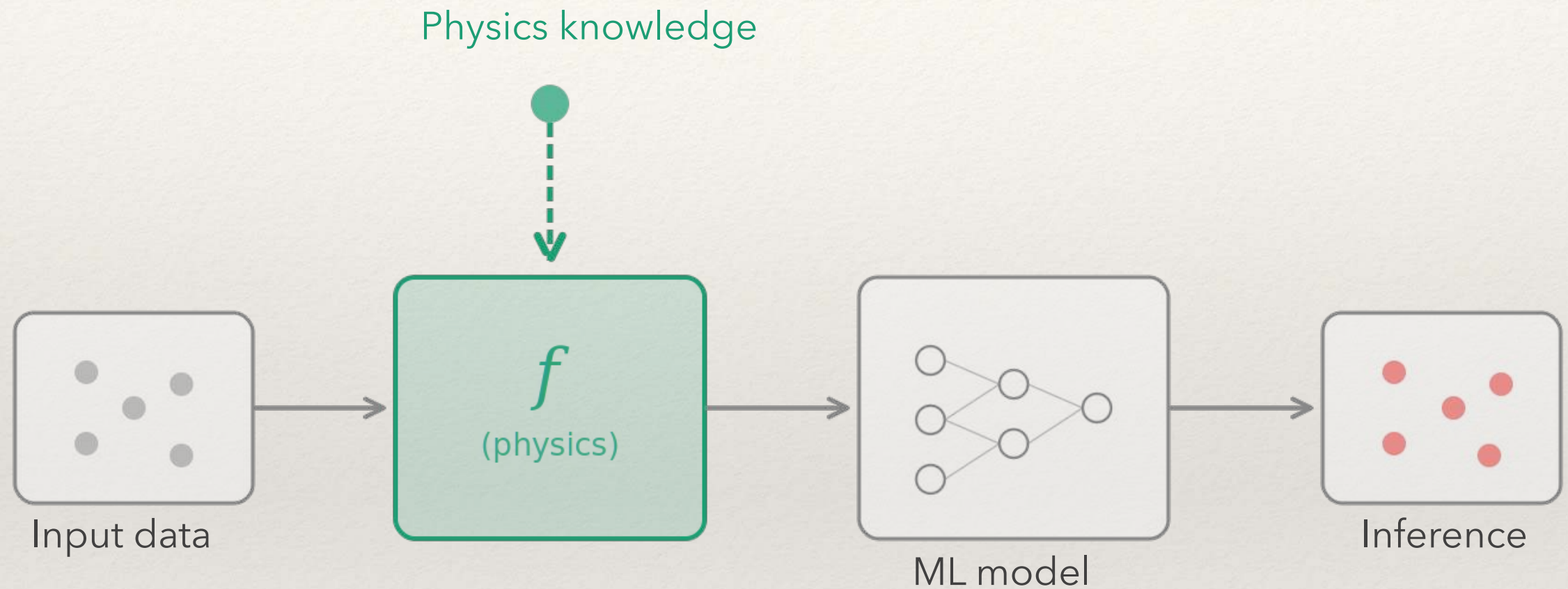
- Often solving differential equations is not sufficient, they need to be aided with some machine learning techniques and this leads to the idea of physics inspired machine learning

This is easier said than done

Physics as hint  Physics as geometry of the solution space

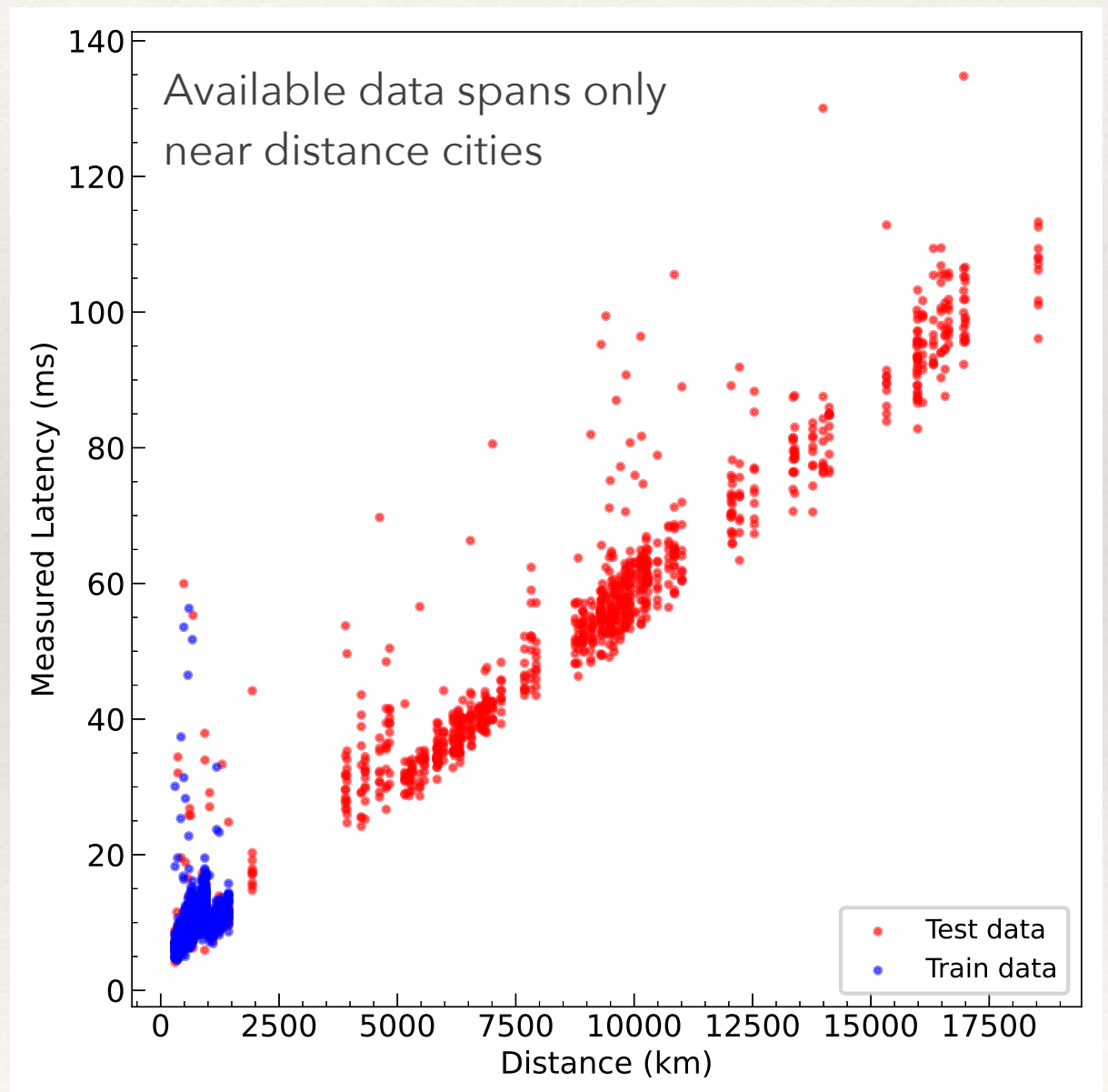
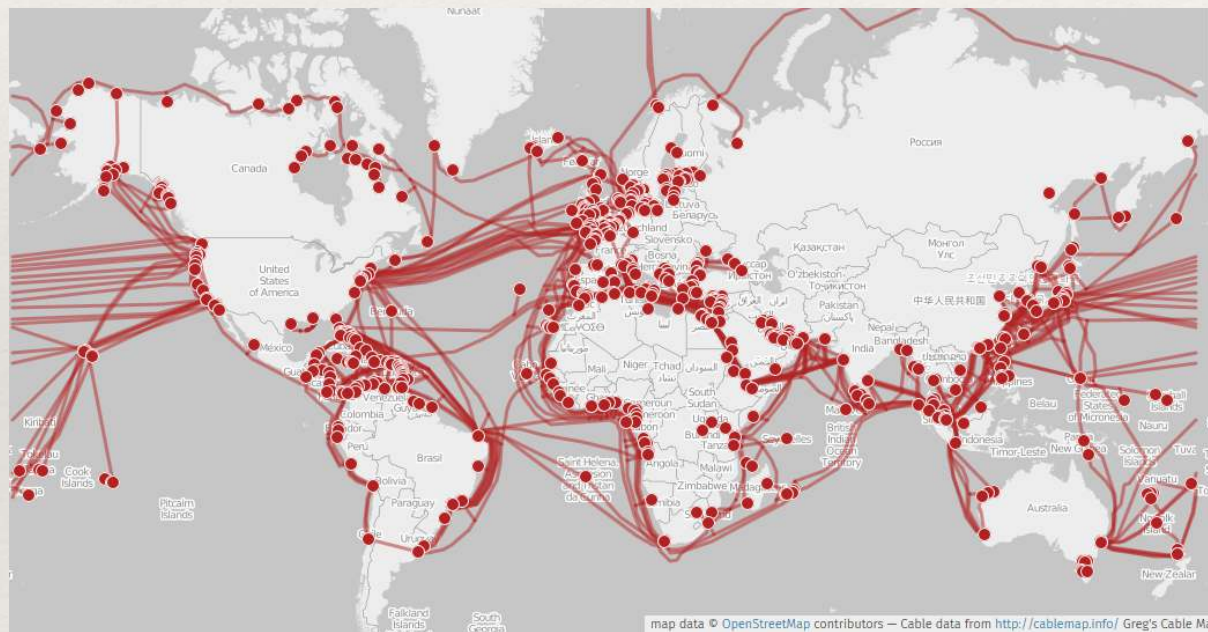
Feature engineering  Physics as inductive bias  Physics as structure

Physics informed feature engineering



Physics informed feature engineering

- Case: We've been given data of time taken for internet signals to reach from point A to point B. We know the geographical distance between the two cities but not the length of the fibre. We are supposed to flag anomalous network communication.



Physics informed feature engineering

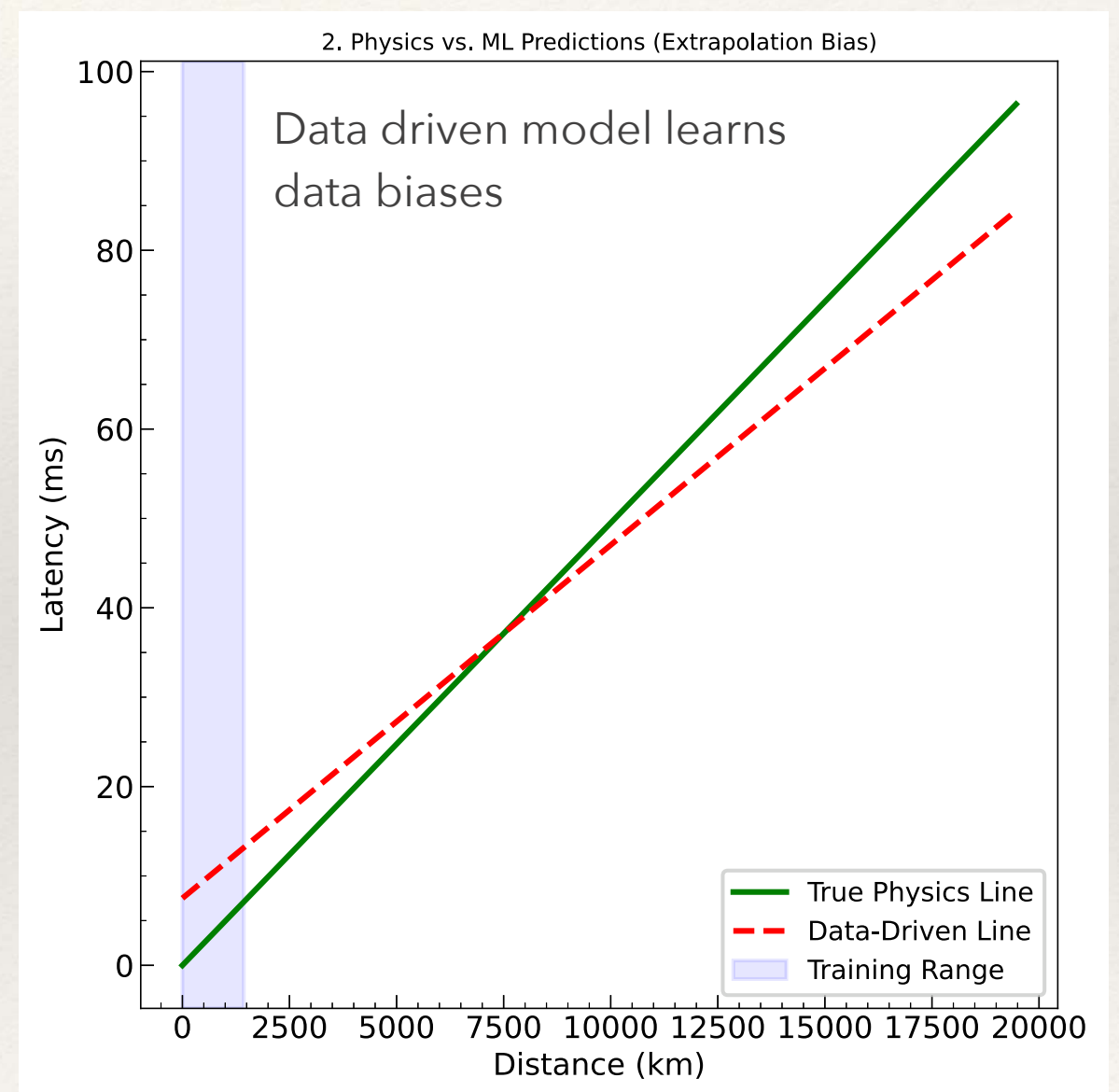
Physics model (no ML)

$$\frac{dx}{dt} = v \Rightarrow \text{latency}_{\text{pred}} = \frac{X_{\text{city}_1} - X_{\text{city}_2}}{\text{speed of light in fibre}}$$

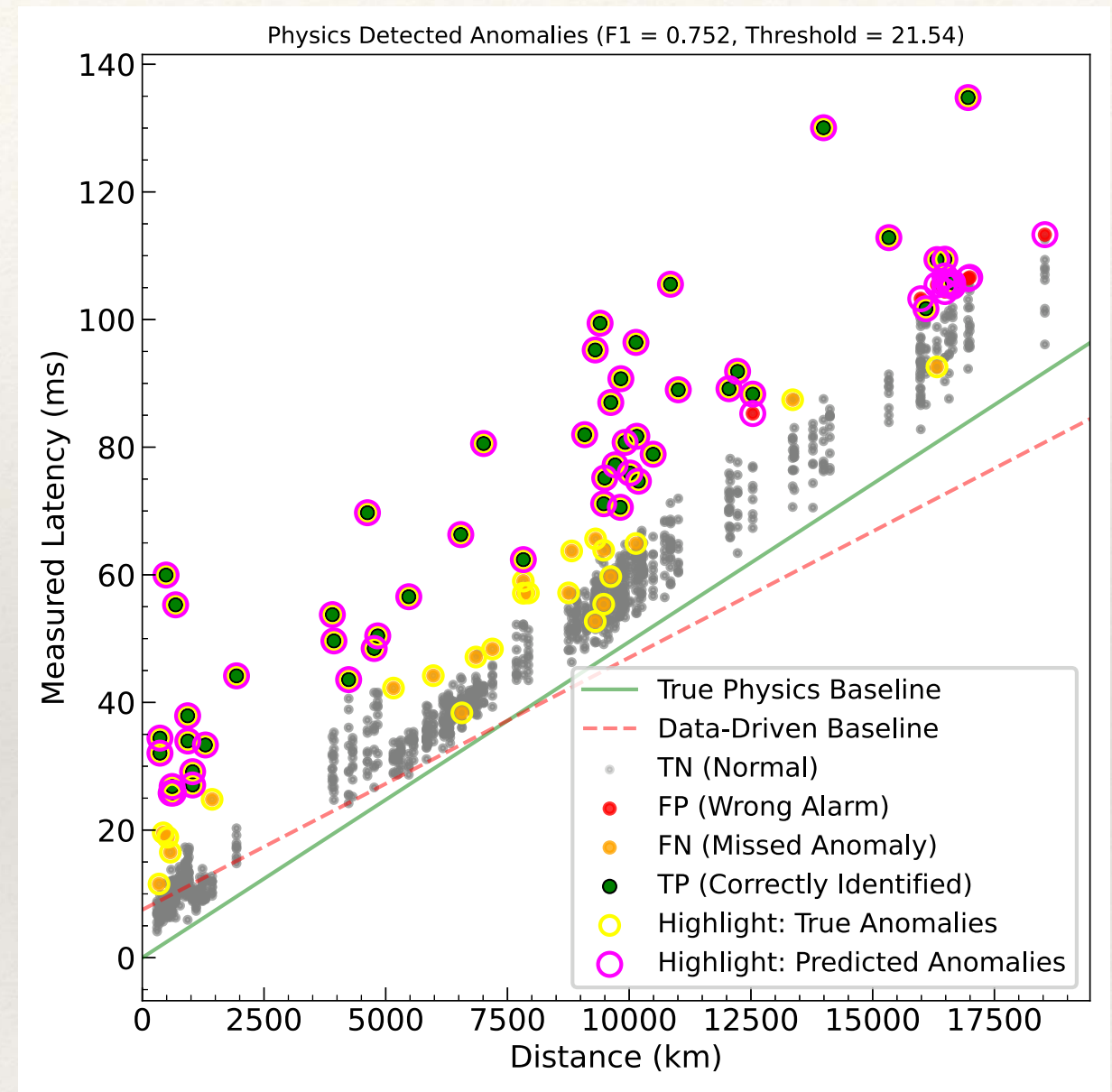
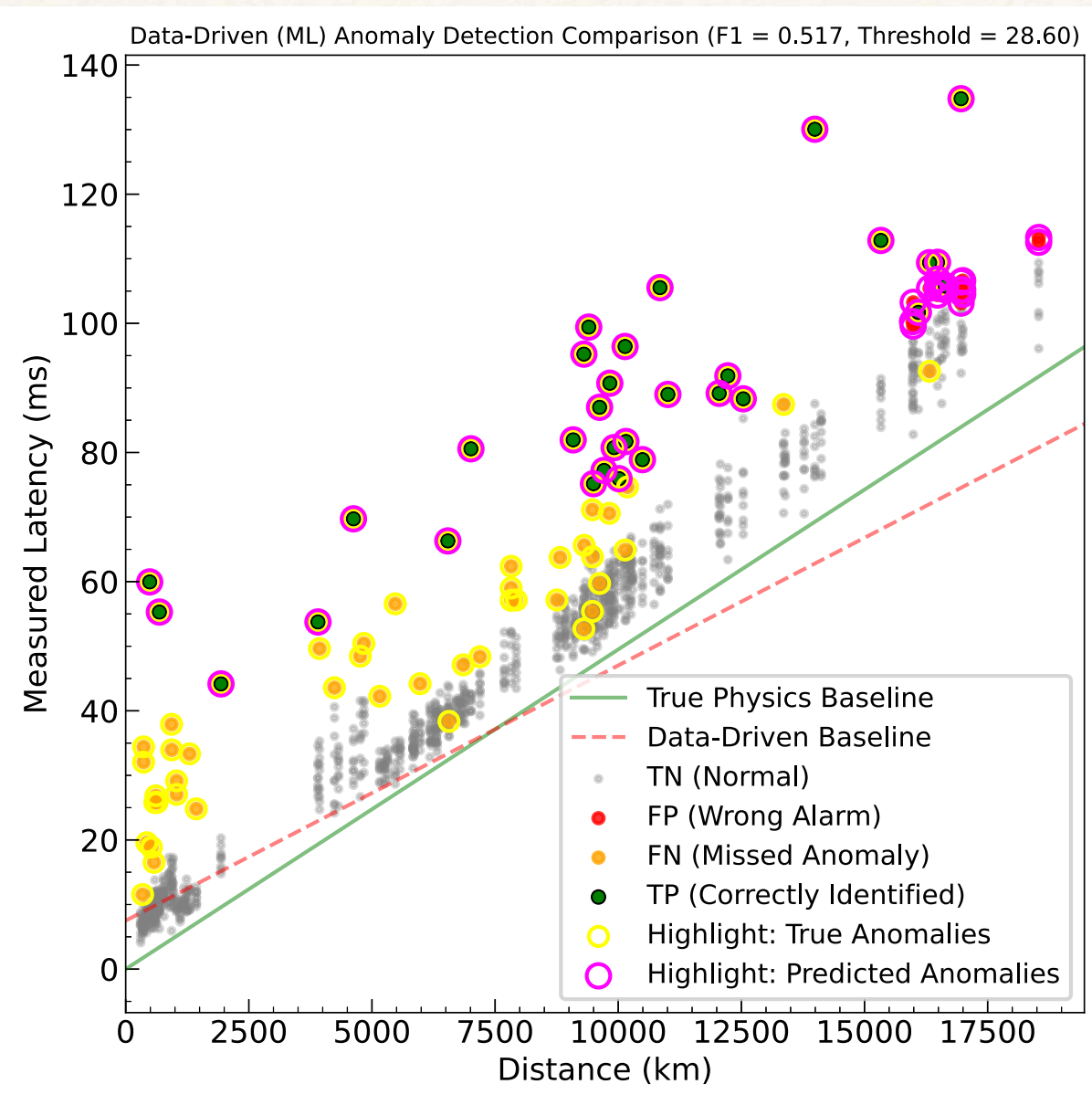
If $\text{latency}_{\text{pred}} < \text{latency}_{\text{measured}}$ flag anomaly
Find quantitative estimate of this inequality
via threshold tuning of precision and recall

ML model

Fit linear regression, find optimal
anomaly detection threshold by tuning
of precision and recall



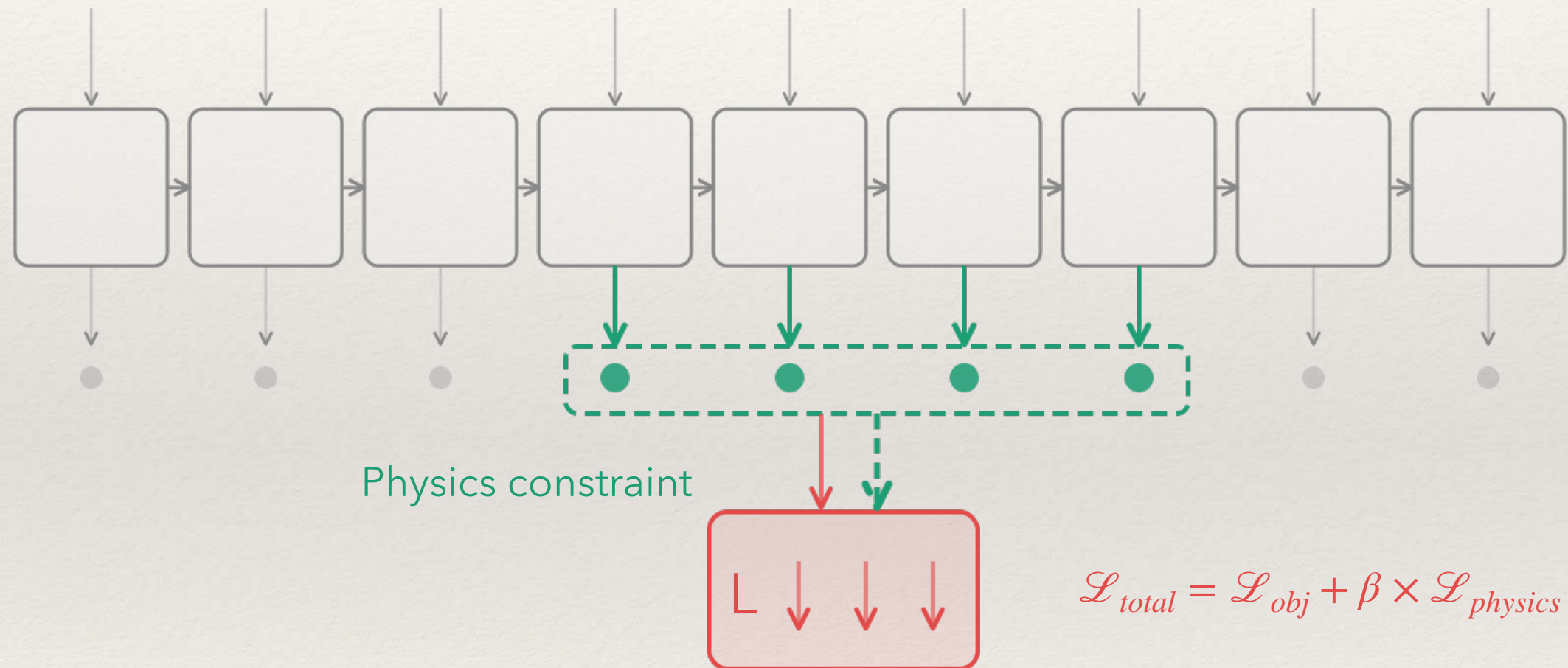
Physics informed feature engineering



- Learn more at: <https://physics-informed-latency-pred.streamlit.app/>

Physics as inductive bias

Input data

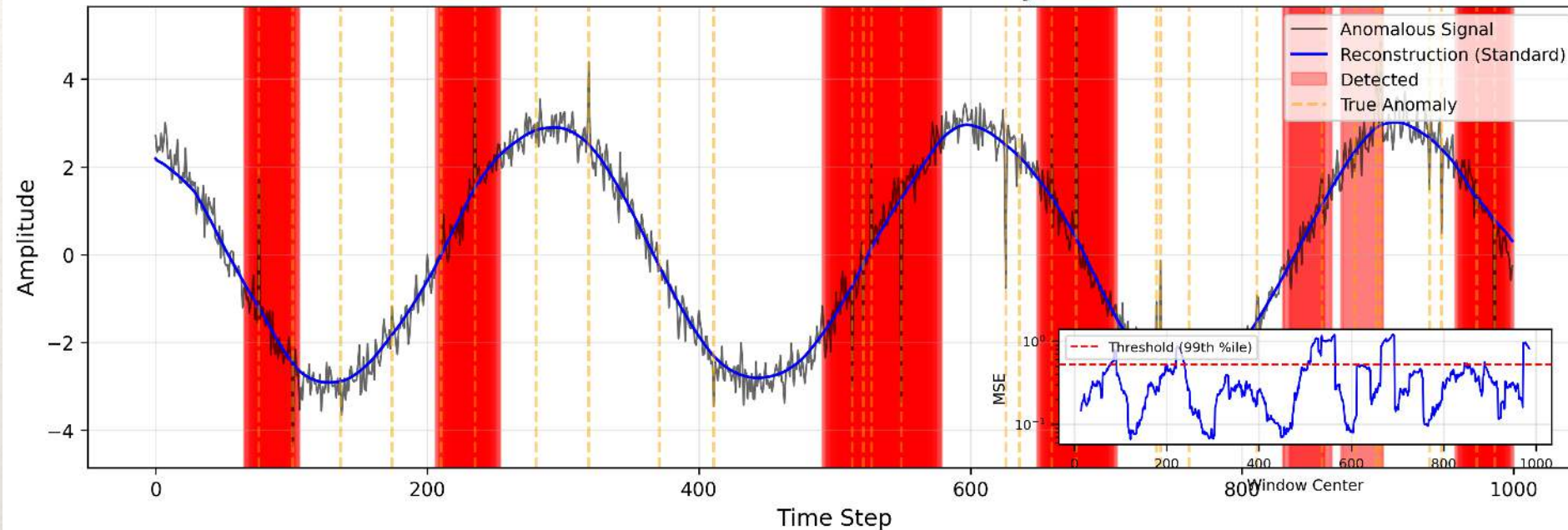


$$\mathcal{L}_{total} = \mathcal{L}_{obj} + \beta \times \mathcal{L}_{physics}$$

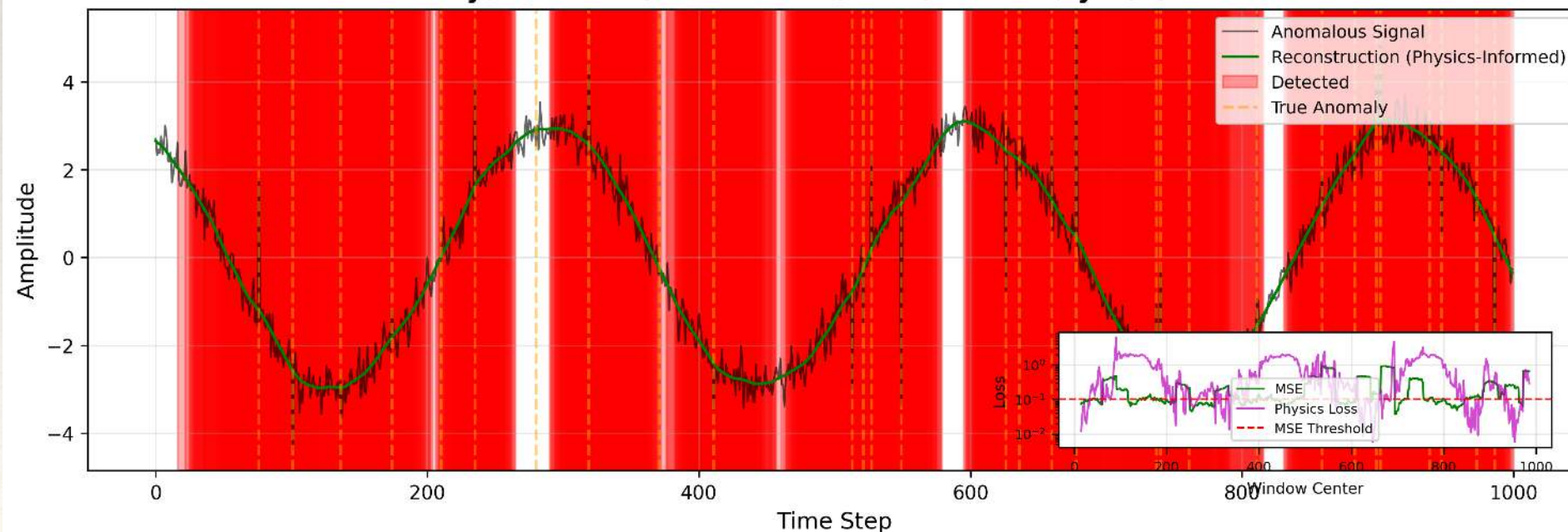
$$\mathcal{L}_{phy} = \frac{d^2 y_{recon}}{dt^2} + \omega^2 y_{recon}$$

Physics as inductive bias

Standard LSTM Autoencoder - Anomaly Detection

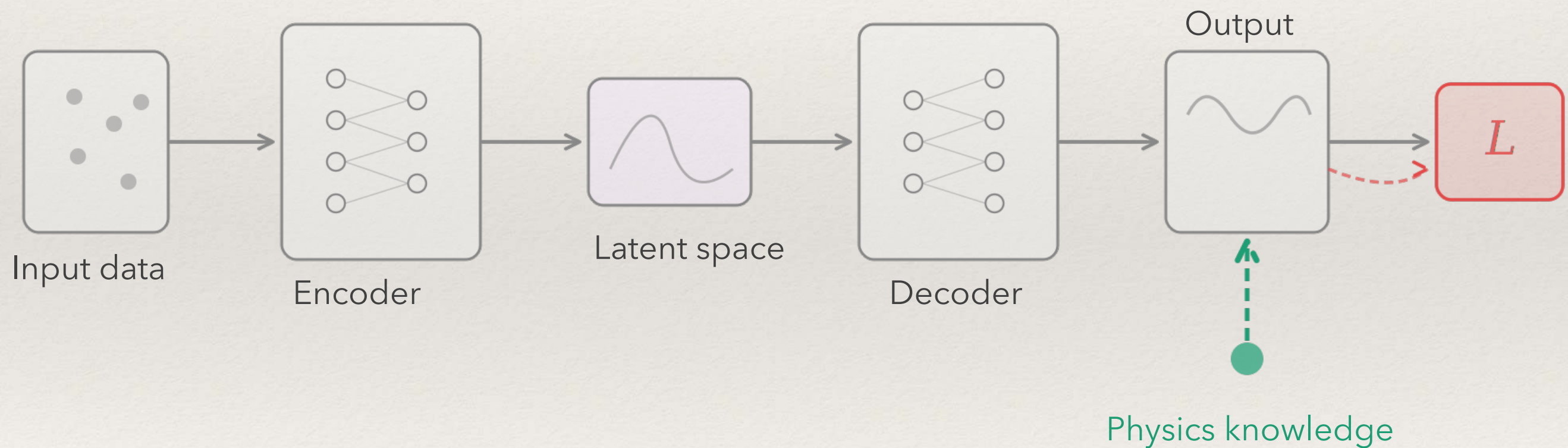


Physics-Informed LSTM Autoencoder - Anomaly Detection



- Both data driven and physics constrained LSTM reconstruct the oscillator well, but physics constrained LSTM identifies close to 100% of anomalies
- Real life caveats: data drift, noise, missing data, unknown frequency, unstable derivatives

Physics as inductive bias



Physics as inductive bias

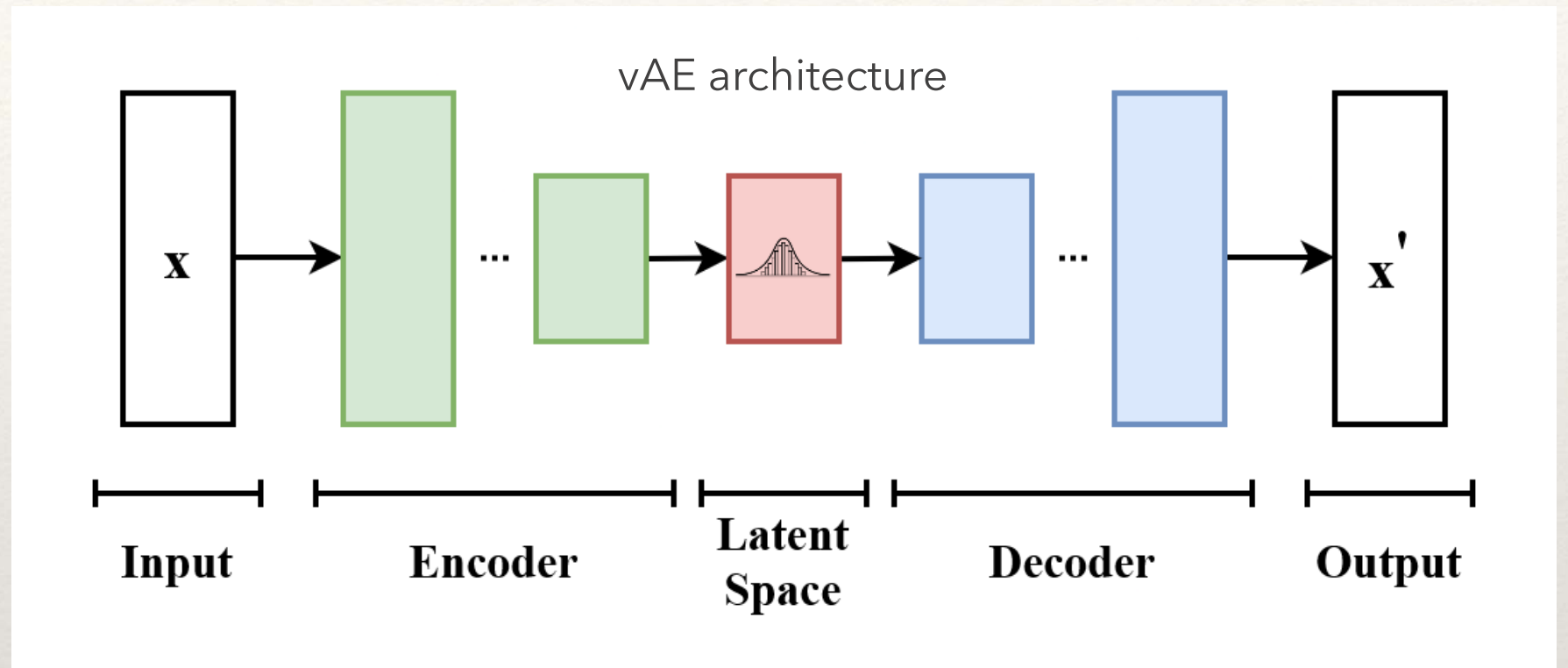
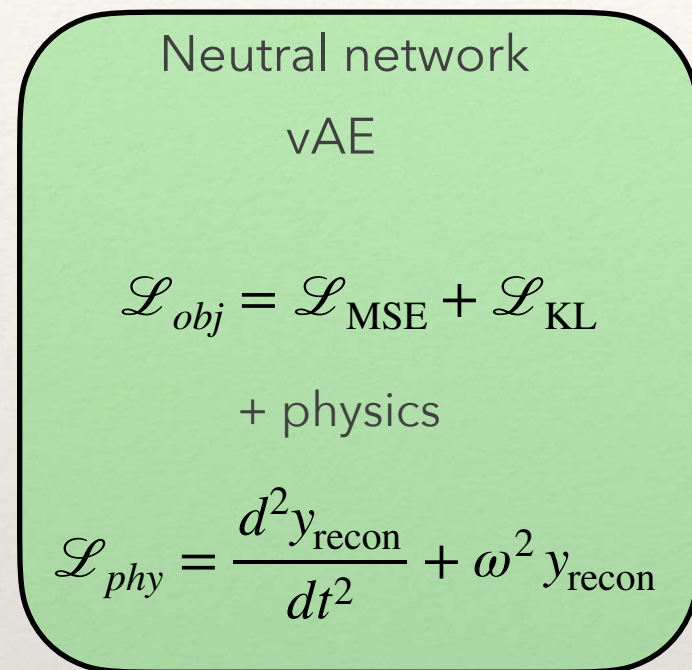


Image source: wiki

β : additional hyper-parameter, needs tuning

- Beta introduces a competing gradient signal (A problem known all too well)
 - May lead to training instability, especially in vAE where mode collapse (i.e. ignoring latent space) is already an issue

Physics as inductive bias

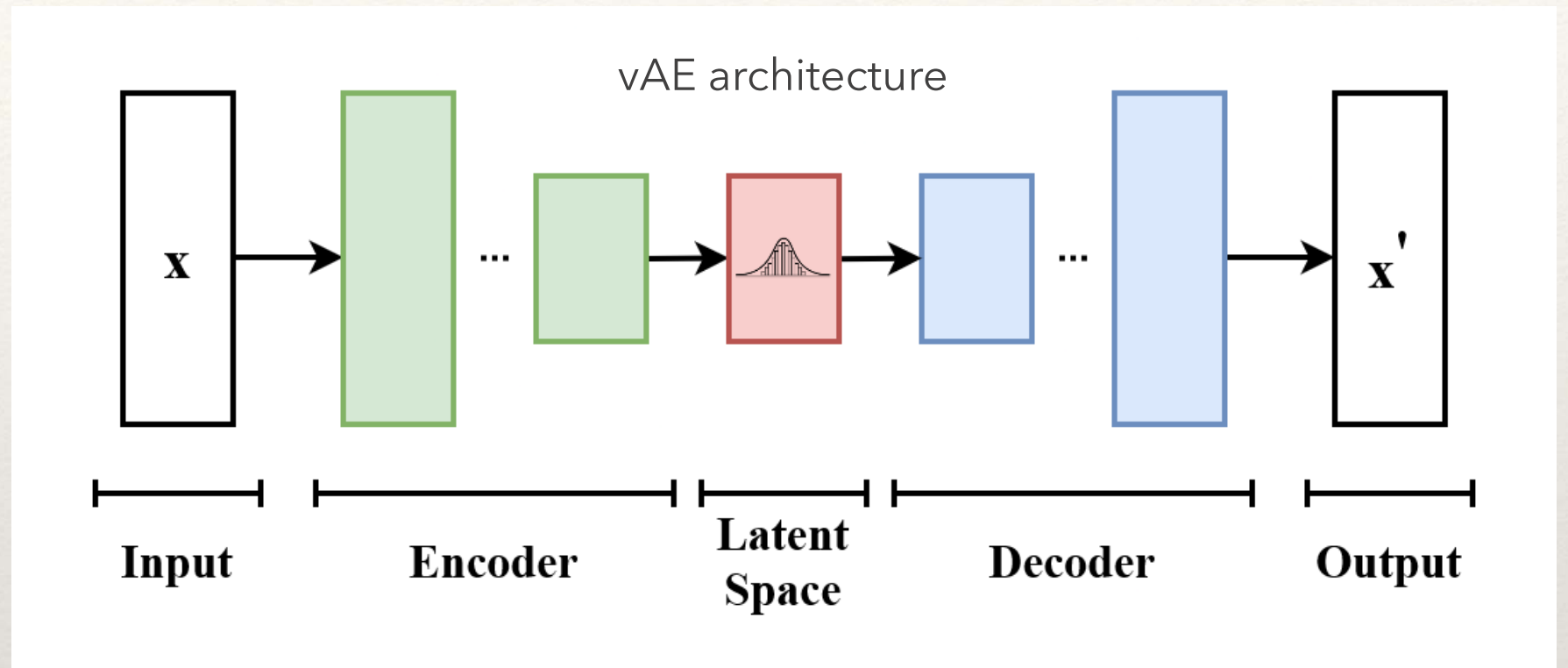
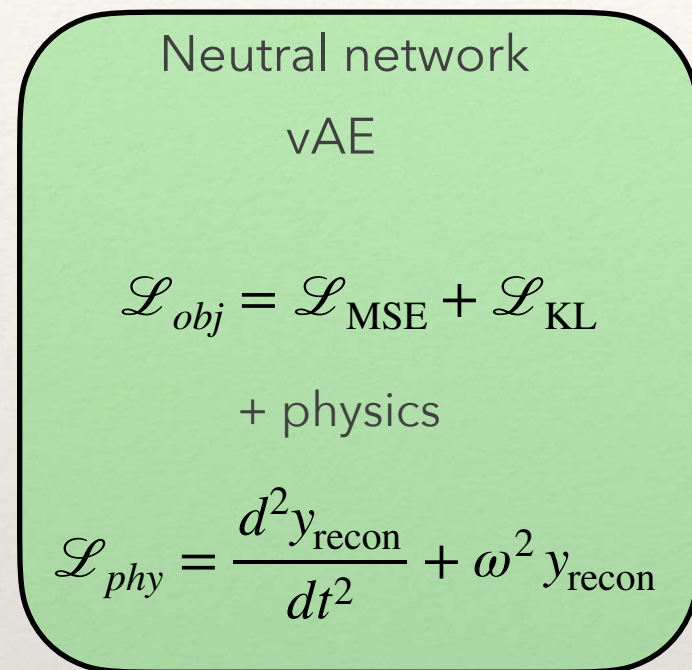


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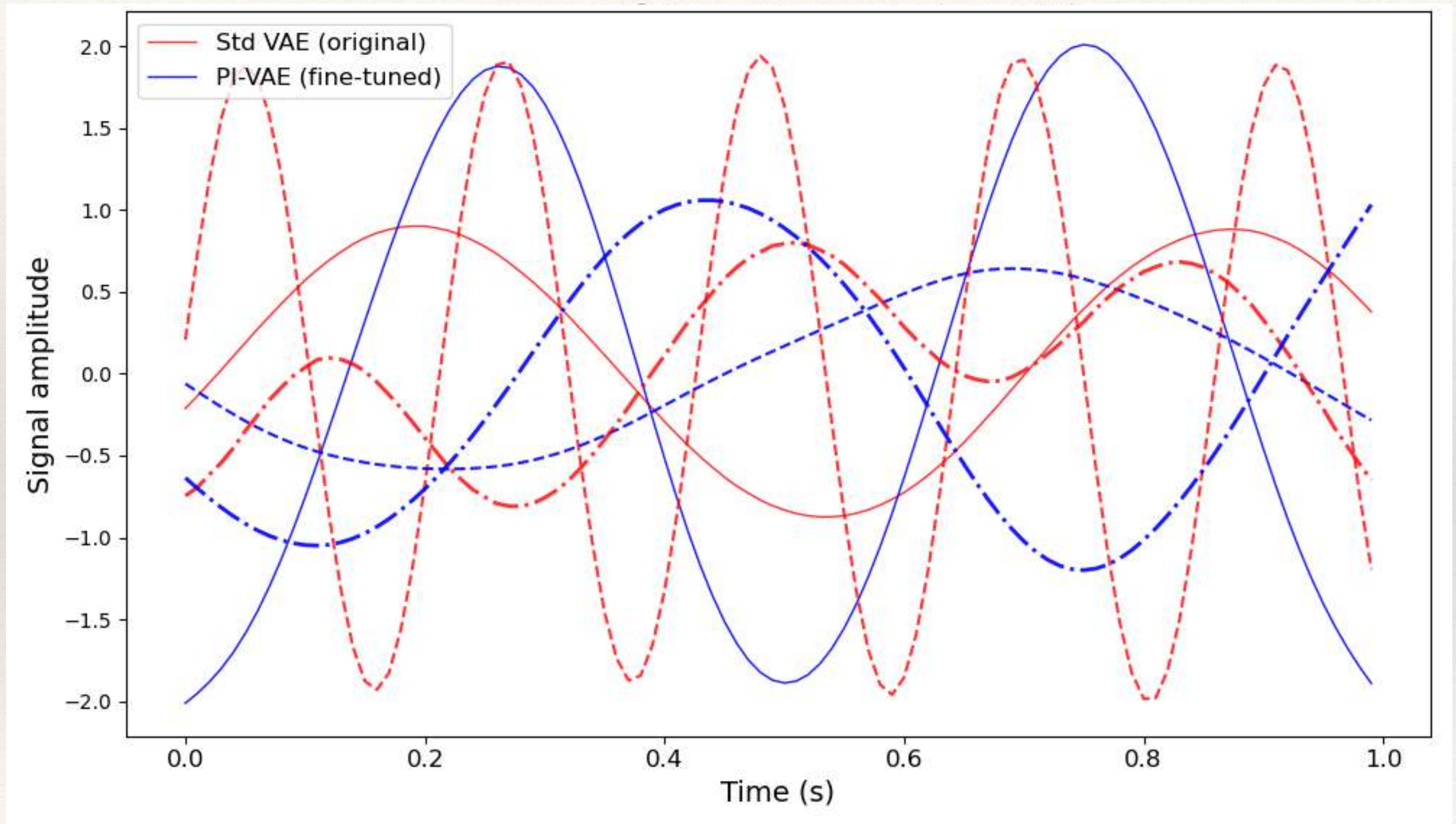
$\Rightarrow$

$$\mathcal{L}_{total} = \mathcal{L}_{obj} + \beta \times \mathcal{L}_{physics}$$

β : additional hyper-parameter, needs tuning

- Beta introduces a competing gradient signal (A problem known all too well)
 - May lead to training instability, especially in vAE where mode collapse (i.e. ignoring latent space) is already an issue

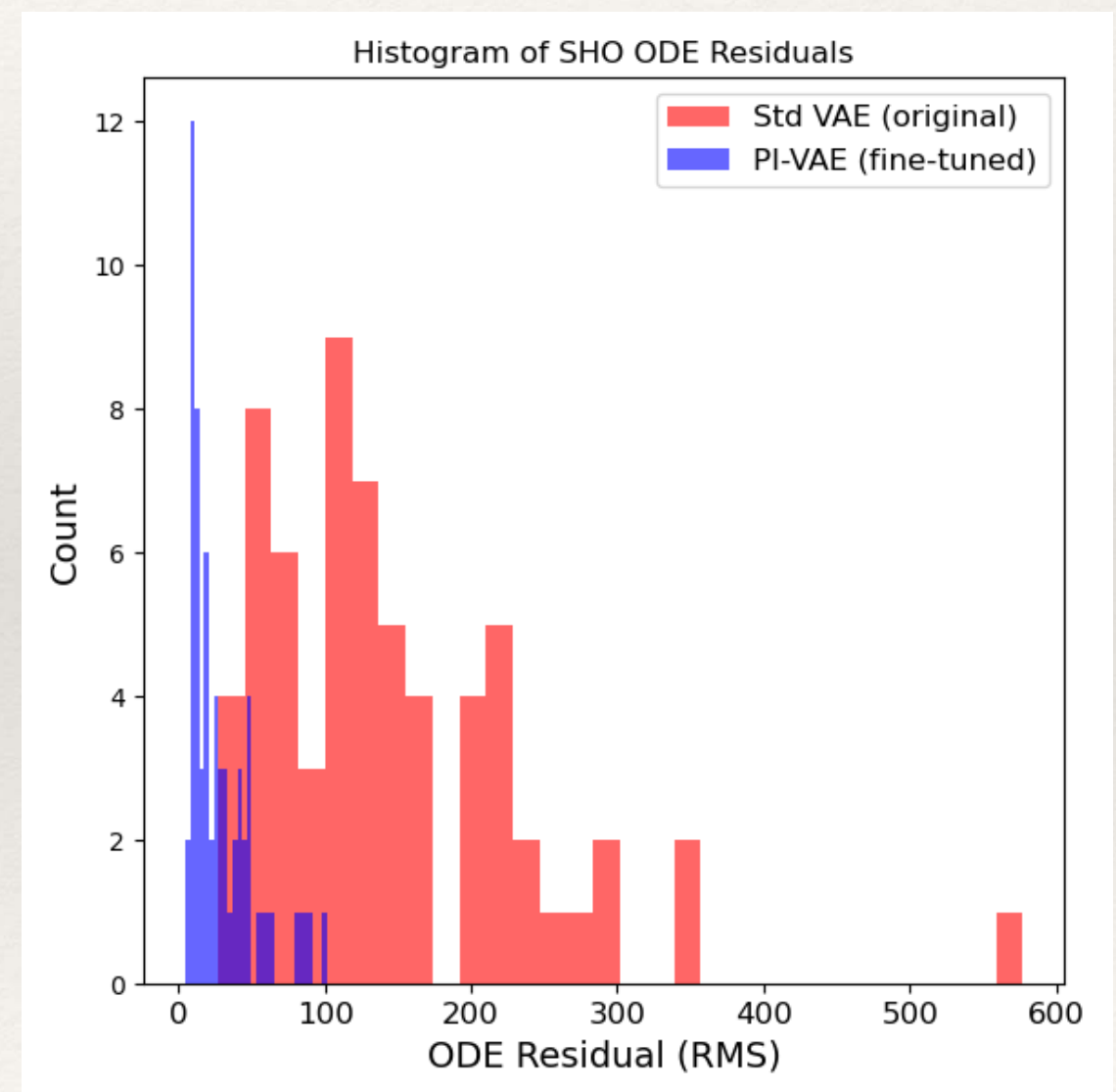
Signal generation



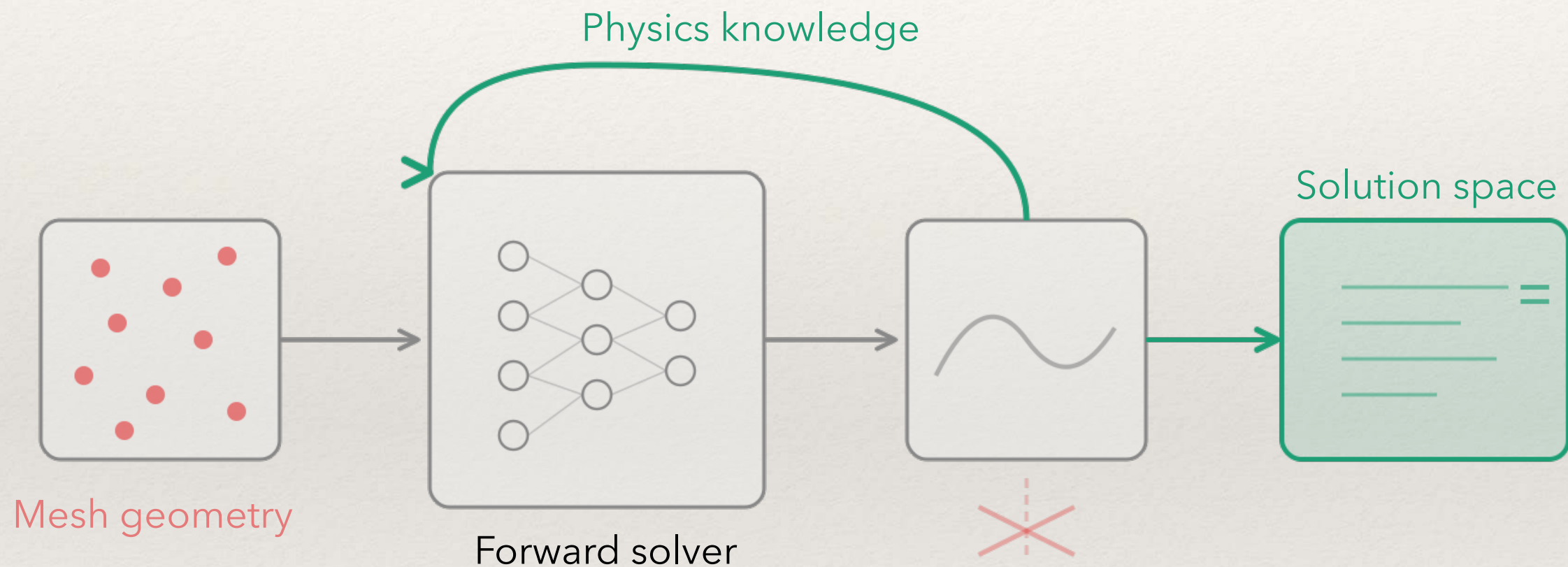
- PI-VAE has a more stable performance in signal generation

Physics fidelity

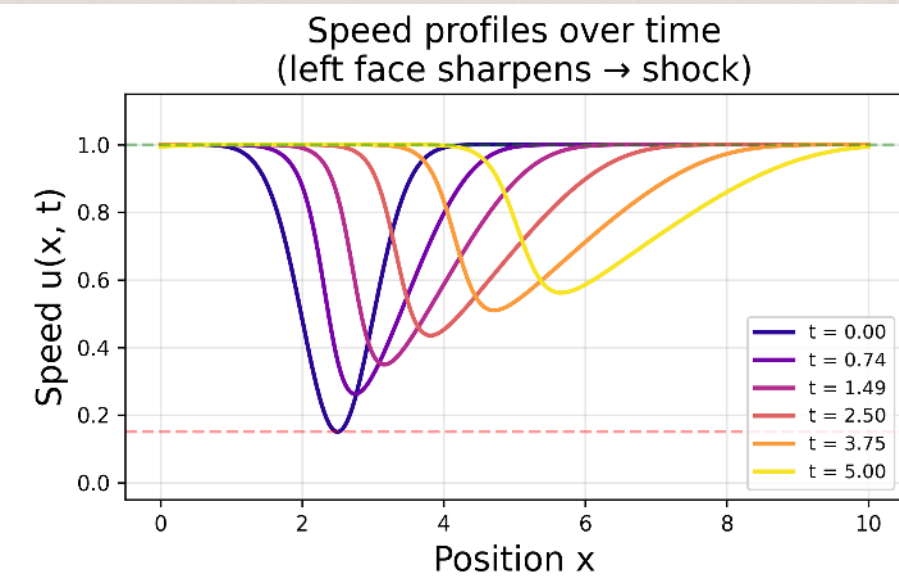
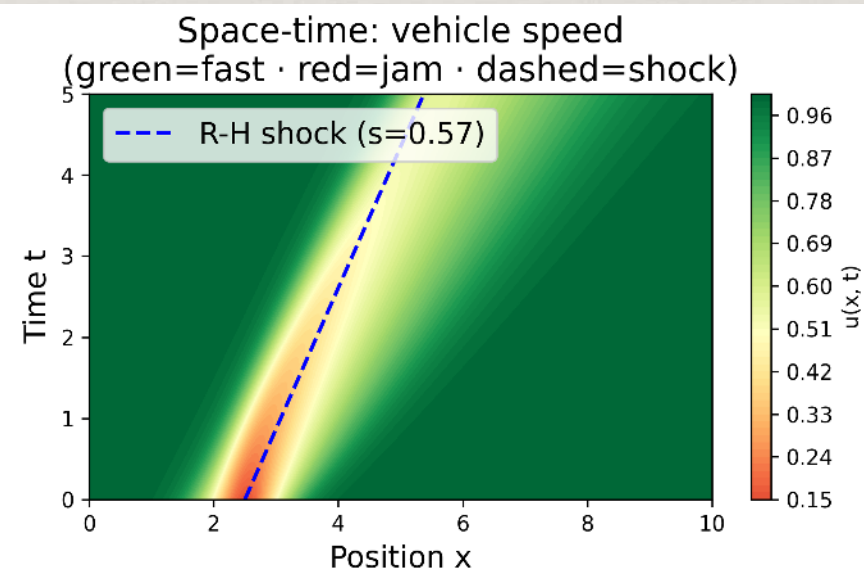
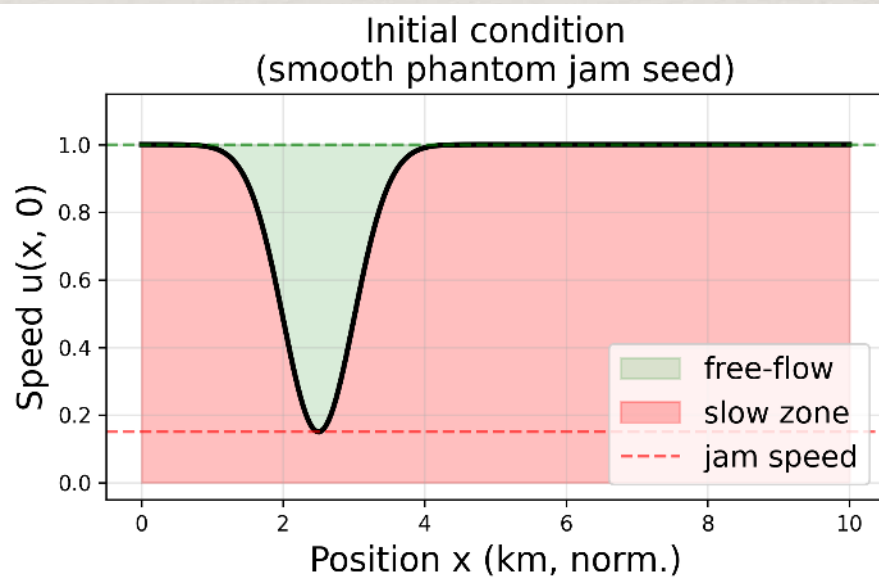
- Physics informed signal generation adheres to physically meaningful trajectories
 - PI-VAE cuts simulation errors by 80%, letting us certify designs faster and beat competitors to market.



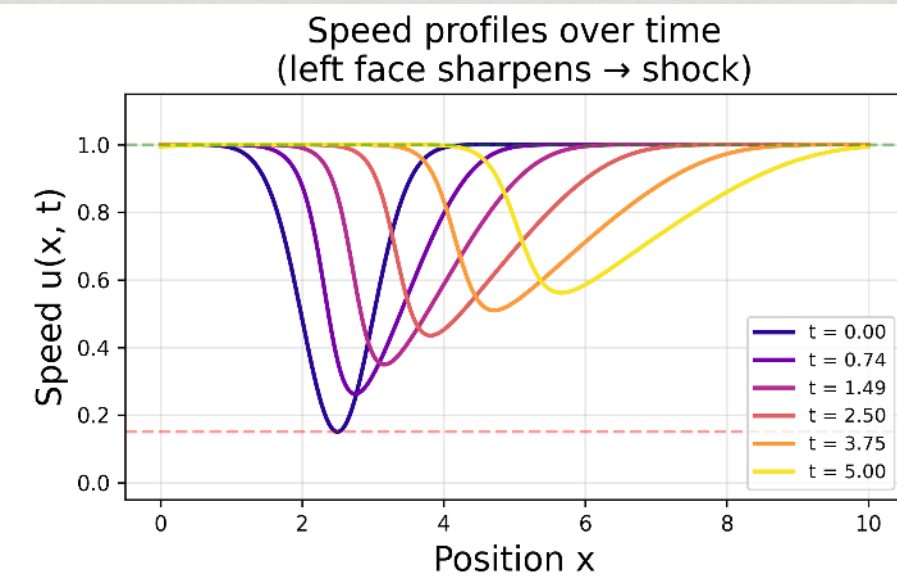
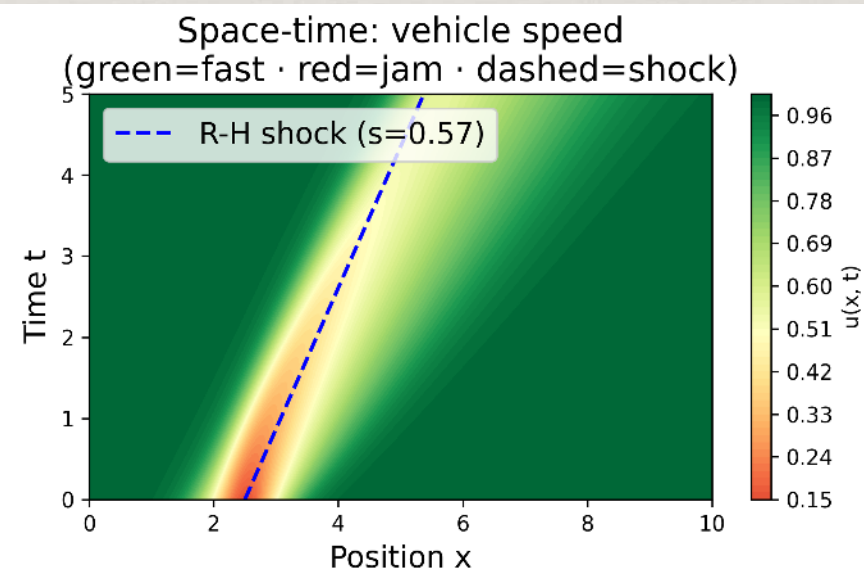
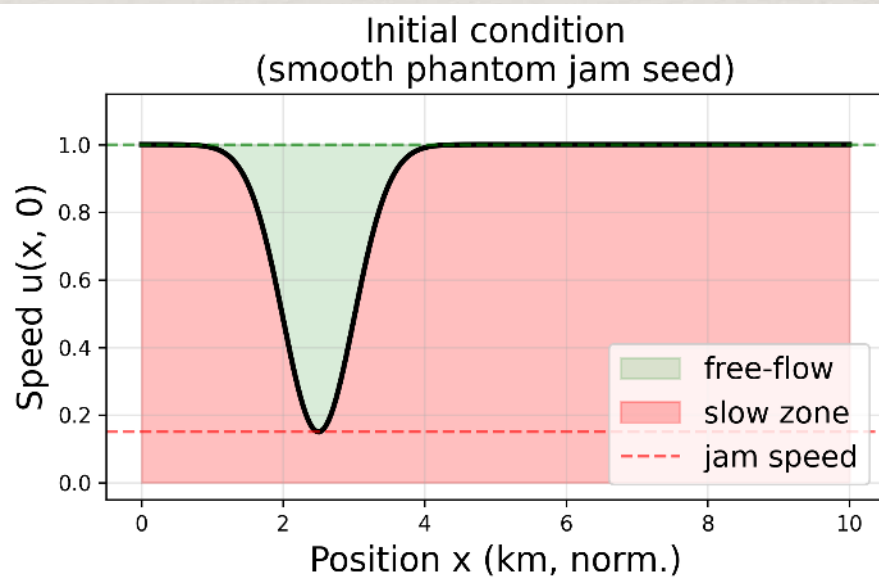
Physics as the training signal



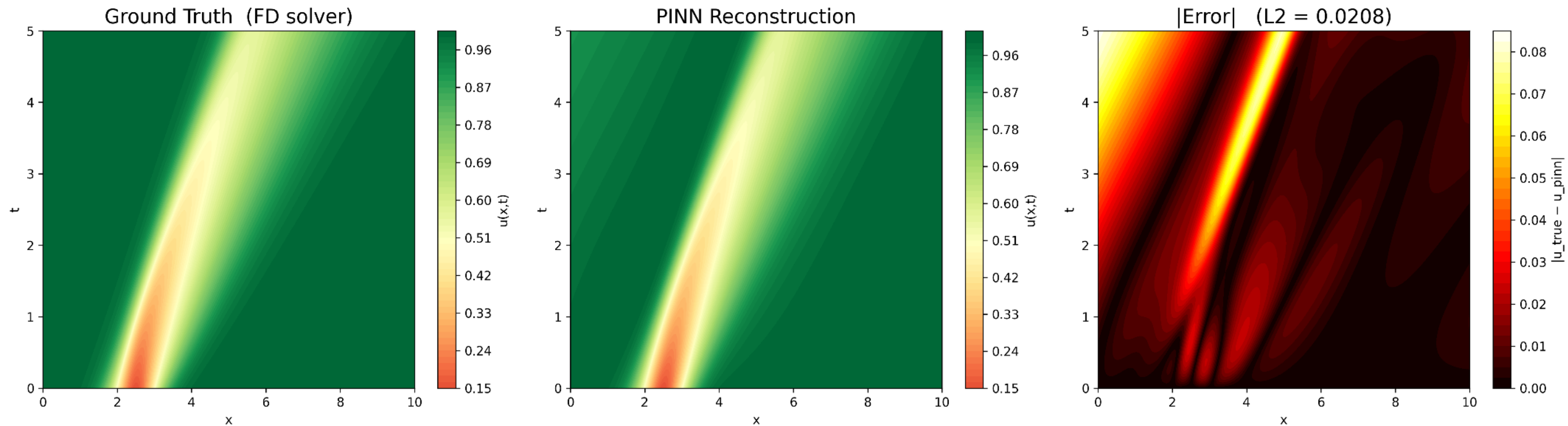
Physics as the training signal



Physics as the training signal

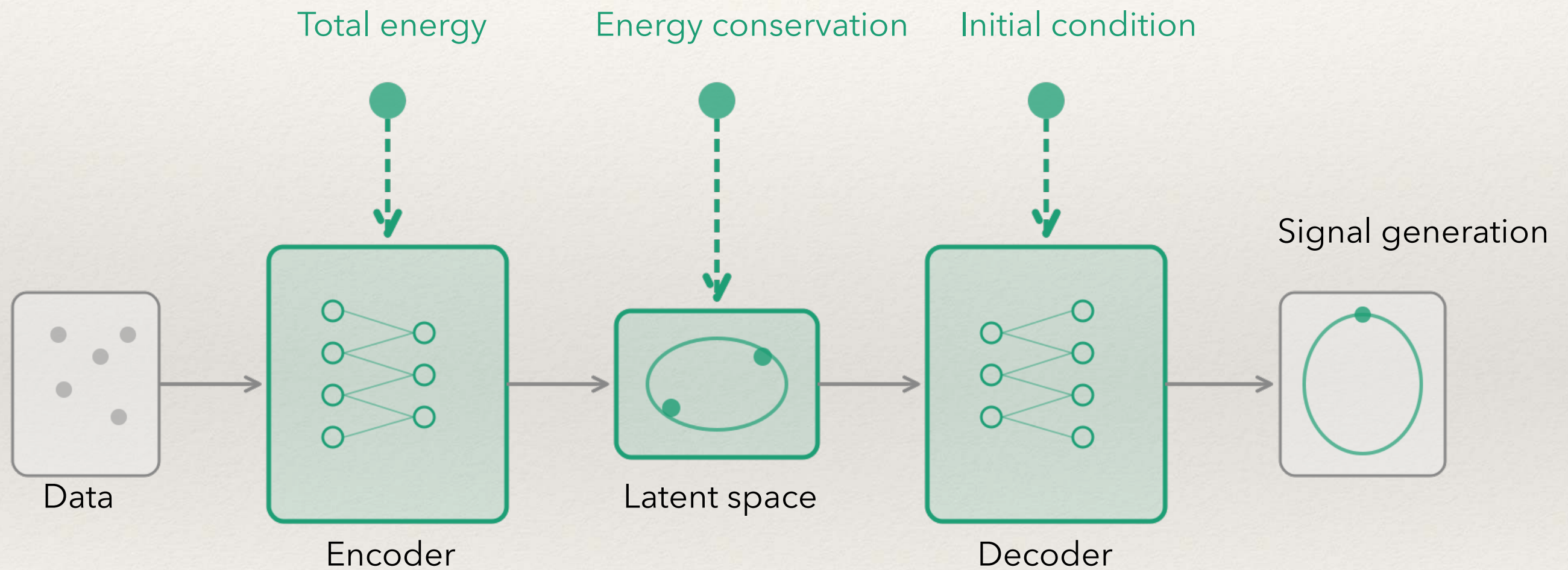


Physics as the training signal

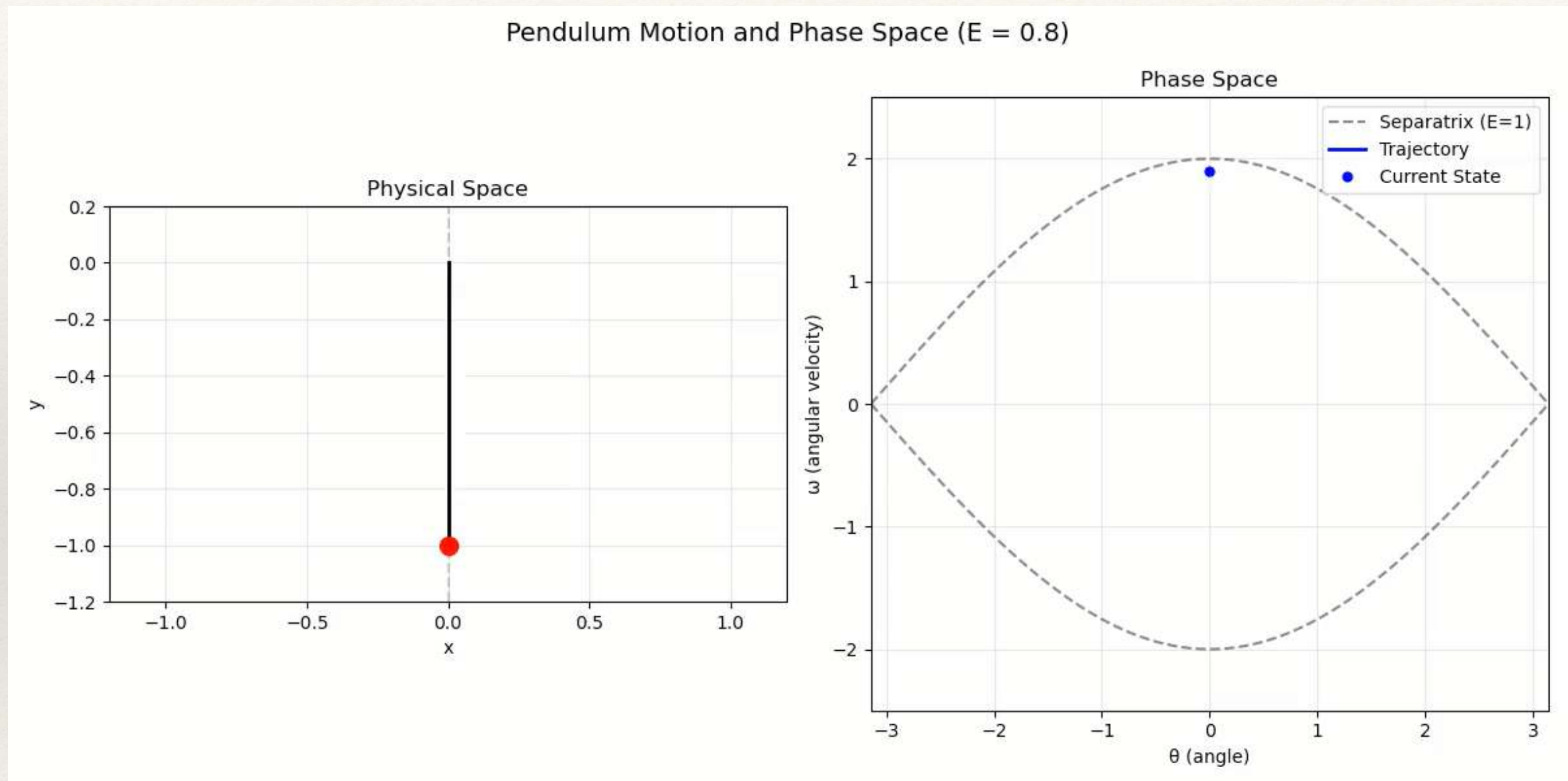


- PINN reconstruction does not involve any data, but computes derivatives in differential equations with autograd
- Error concentrate at the discontinuity, an open problem in such approaches (PINNs)

Physics as the training signal

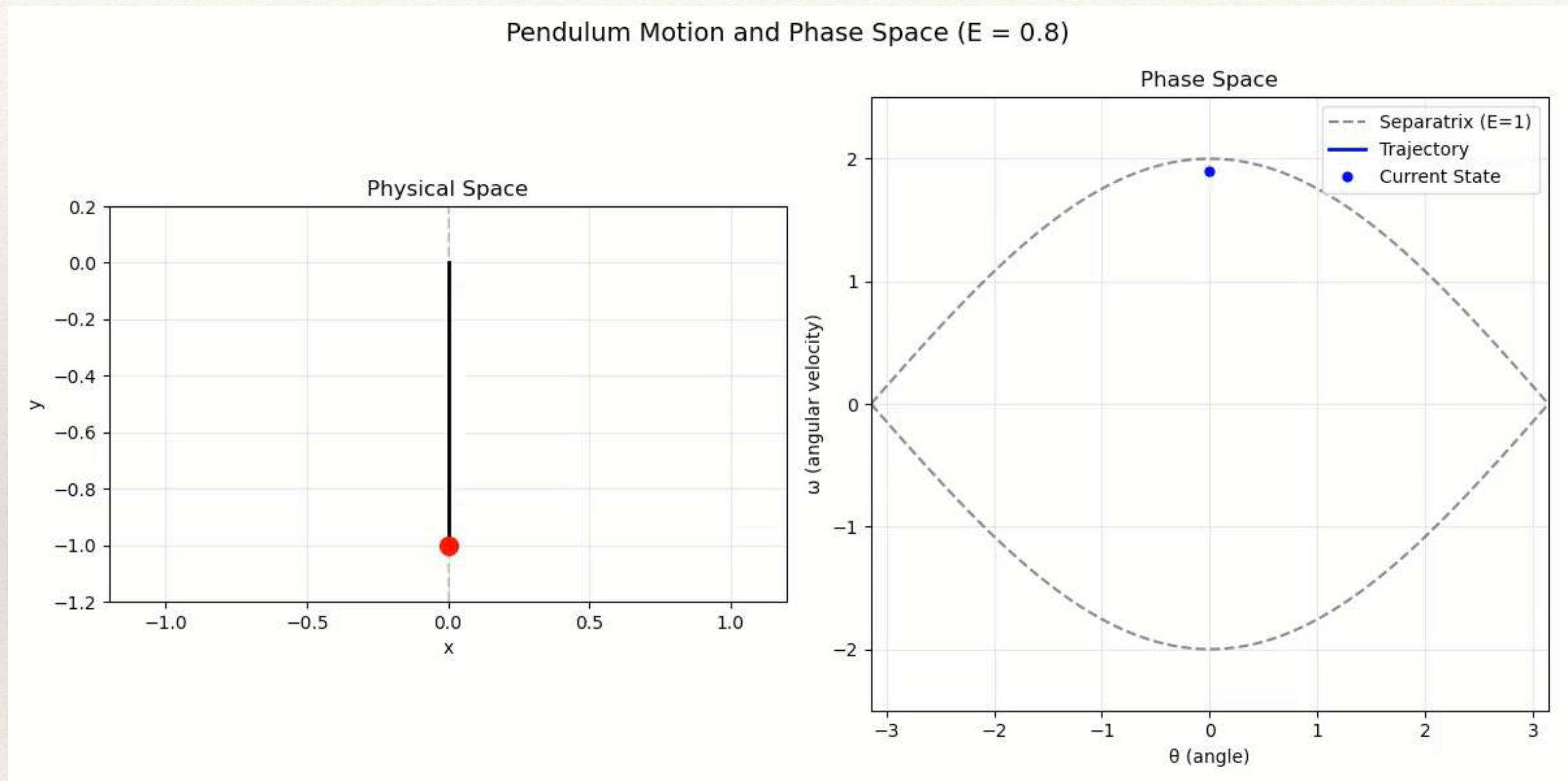


Physics as structure, not penalty



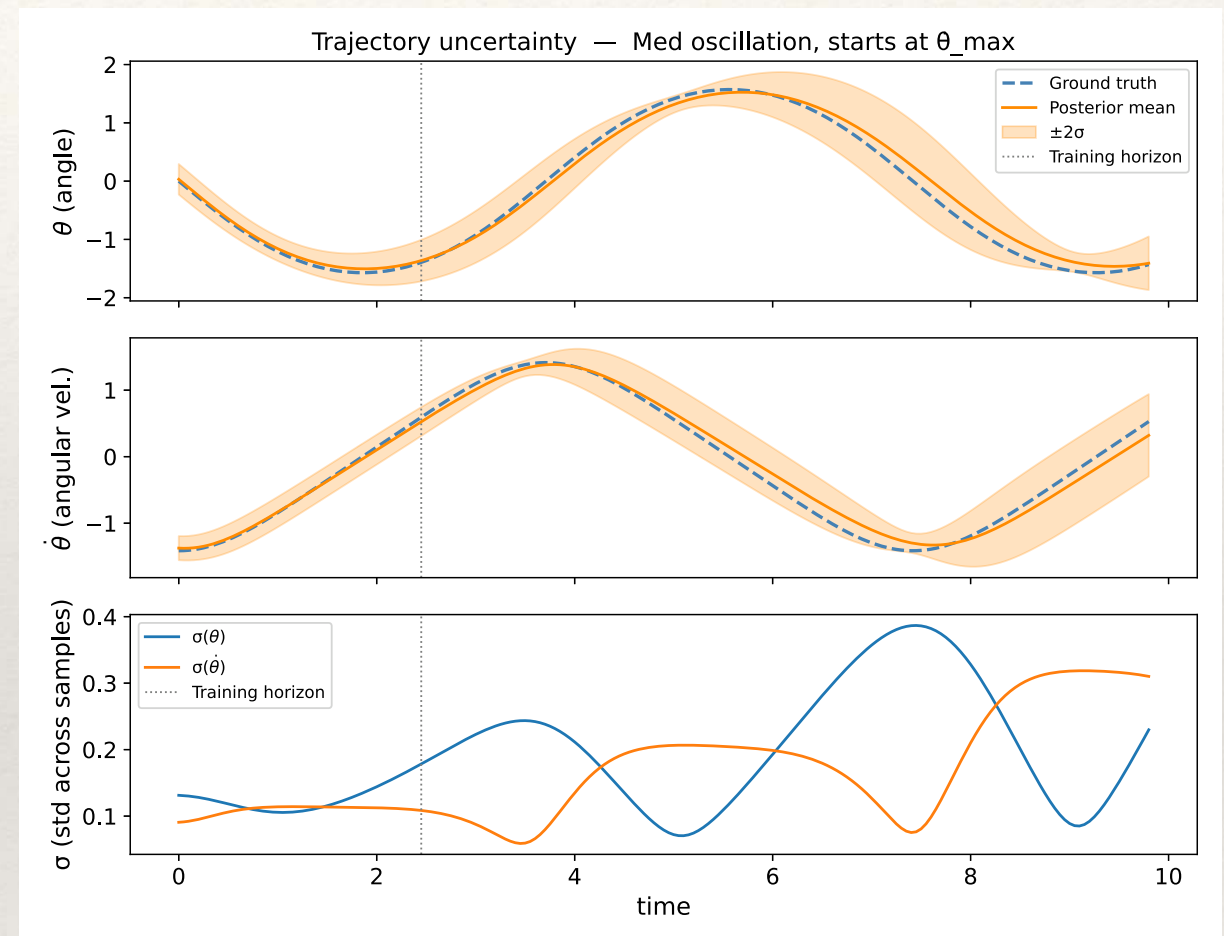
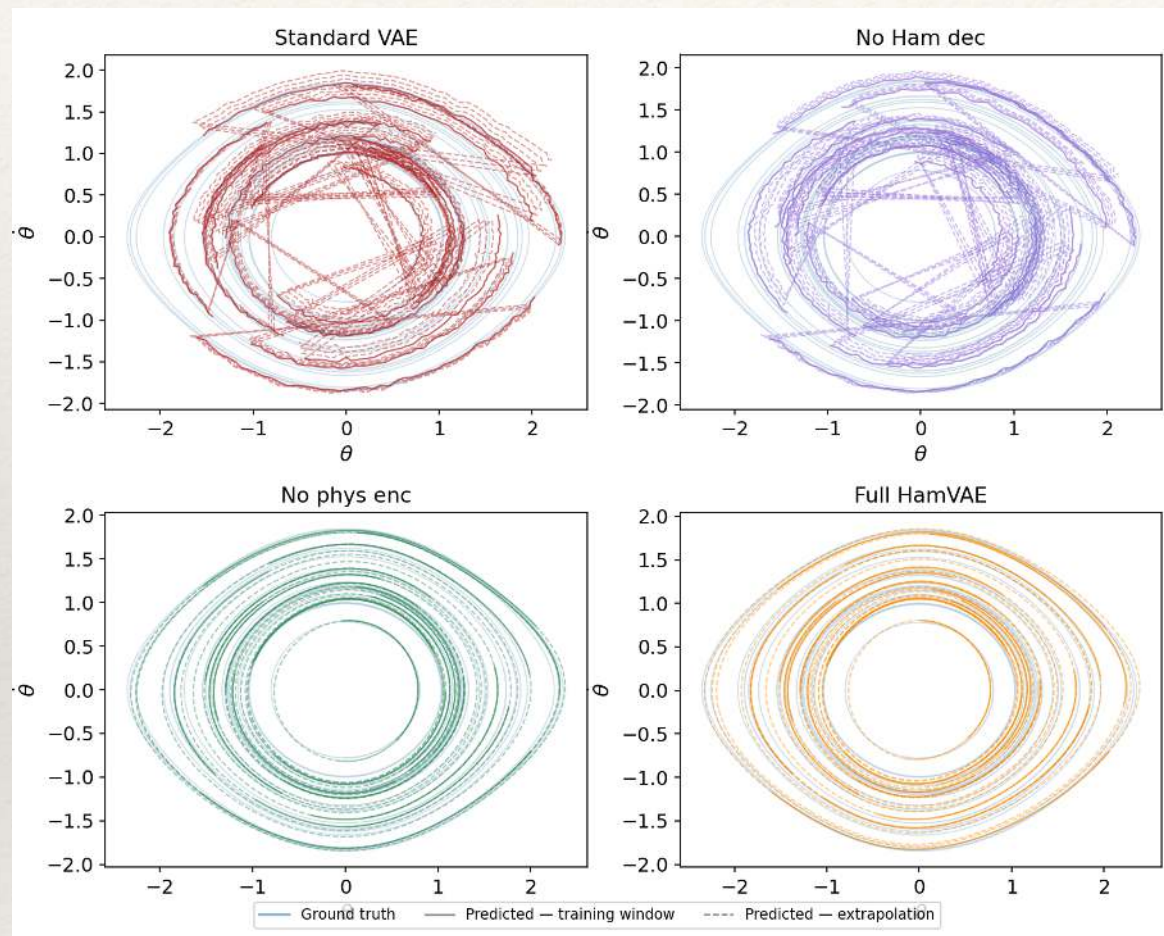
- Unlike simple harmonic oscillator, there is no Taylor expansion here

Physics as structure, not penalty



- Unlike simple harmonic oscillator, there is no Taylor expansion here

Physics as structure, not penalty



- Clear generalisation with uncertainty under control

Conclusions

When data does not tell the whole story

- When data collection costs a borehole, a crash test, or a reactor
- When your model needs to work beyond the training distribution
- When a wrong answer is expensive

Physics is the cheapest regulariser you will ever have.

